When canonical covariance localisation fails

Exact finite-ensemble counterexamples, a sharp leading-order criterion, and the Snyder–Hakim test case

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Abstract

In plain terms. Weather and ocean forecasts are corrected with observations by an ensemble Kalman filter, which estimates forecast uncertainty from a small set of forecasts and then *localises* that noisy estimate. Snyder and Hakim (2022) proposed a special choice of variables, *canonical coordinates*, and conjectured that it is the best one to localise in. We show that whether this is true depends on how analysis error is scored: it fails for error measured in the ordinary physical variables, which Snyder and Hakim used in their experiments, and it holds at leading order for error measured relative to the forecast uncertainty.

Setting. One analysis with an independent Gaussian ensemble of size N, an oracle (population) prior covariance used to build every basis, and deterministic Schur-product tapers. Canonical coordinates whiten the prior and observation errors and then diagonalise the whitened observation operator by a singular value decomposition.

Physical squared error. With one scalar observation that is not an eigenvector direction of the prior covariance, an exact first-variation formula shows that, at every $N \geq 2$, a sufficiently small rotation followed by complete localisation (keeping only sample variances) beats every deterministic symmetric canonical taper with unit observed diagonal. Allowing a deterministic variance rescaling in the rotated basis also yields an estimator whose risk is strictly below the infimum over all canonical tapers with any observed-variance factor. Two two-dimensional examples are certified in exact arithmetic; in one, the canonical basis is a strict local *maximum* of the risk. With several observations, coupling between an observed and an unobserved canonical direction makes canonical complete localisation non-stationary at every finite N; this finite-N result compares with canonical complete localisation only, not with all canonical tapers.

Large ensembles. For a general observation matrix we prove a uniform risk expansion whose leading basis-dependent term is $F(U) = \sum_i (u_i^\mathsf{T} B u_i)(u_i^\mathsf{T} A u_i)$. Canonical coordinates minimise F exactly when, in the canonical directions, the loss weights are decoupled from one another and ordered opposite to the observation information. Under prior-normalised (Mahalanobis) loss this always holds, so among complete localisations and unit-diagonal tapers in whitened bases the conjecture is true at leading order. For a scalar observation at finite N, canonical complete localisation is a strict local minimum (up to the irrelevant completion of unobserved directions) if a scalar ζ is positive and is not a local minimum if $\zeta < 0$; $\zeta > 0$ holds in every case computed but is unproved.

Snyder and Hakim's test case. In a reconstruction of their 200-variable example, symmetric (ZCA) whitening or the best basis found removes about 54% of the canonical basis's excess error over the Kalman filter at $N = 8$ (a reduction of 0.074 in MSE relative to the Kalman filter), roughly equivalent to doubling the ensemble; under prior-normalised loss the canonical basis is the best found. Proofs are not machine-verified; exact-arithmetic and floating-point checks are distinguished throughout.

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1 Introduction

1.1 Background in brief

What an ensemble analysis does. Data assimilation combines a forecast (the prior) with new observations to produce an analysis. The Kalman filter does this optimally for linear Gaussian problems, but it needs the prior covariance matrix P, which is far too large to store or propagate in weather and ocean models. An ensemble Kalman filter (EnKF) replaces P by the sample covariance of a modest ensemble of N forecasts, typically tens to a hundred members for millions of variables [12].

Why localise. With so few members, the sample covariance contains large spurious entries: two distant variables appear correlated purely by chance. *Covariance localisation* damps these entries by multiplying the sample covariance entrywise (a Schur product) by a fixed taper that decays with distance [12]. Many variants exist, and several theories choose the taper to minimise a stated error criterion in a given set of variables [7, 18, 20, 23].

What canonical coordinates buy. Localisation depends on the variables in which it is applied. Snyder and Hakim [27] observed that there is a linear change of variables in which the Kalman update decouples completely. First whiten the prior, so that the state has identity covariance, and whiten the observation errors. Then rotate both spaces by the singular value decomposition of the whitened observation operator. In these *canonical coordinates* every true covariance that enters the update is diagonal, so any off-diagonal sample covariance is pure sampling noise. The natural localisation is then *complete*: discard every off-diagonal entry and keep only the sample variances.

1.2 What Snyder and Hakim conjectured

Snyder and Hakim define the best transformation as the one "after which an EnKF using covariance localization" gives "the smallest mean-squared analysis error on average" [27, Section 3]. The localisation "may be chosen according to the transformation" (ibid.). They conjecture that canonical coordinates (their Eq. 8) attain this optimum, "but do not prove" it [27, Sections 1 and 7]. Their localisation in canonical coordinates zeroes the off-diagonal entries of both the state–observation and the observation–observation sample covariances [27, Section 3.2]; for a scalar observation this coincides with complete localisation of the state covariance, which is the convention used here (Section 5 states the convention for several observations).

Three features of their paper shape our comparison.

- *The loss.* Their experiments score the mean squared error of the analysis mean in the original variables, normalised by the Kalman-filter value [27, Section 4, Figs 3 and 5]. They also say that other optimal-localisation theories are tied to a choice of variables, whereas their emphasis is "independent of those choices" [27, Section 1]. Physical mean squared error is *not* invariant under changes of variables, so the conjecture needs a loss to be well posed.
- *The only competitor they tested is a rotation.* Their alternative transformation projects the state onto the eigenvectors of P (their Eq. 15). After whitening, this is an orthogonal rotation of the canonical basis, so their Fig. 5 is an experiment inside the competitor family studied here. It favoured canonical coordinates narrowly; they add that they "do not attempt a general answer" [27, Section 4].

- *The oracle setting.* The transformation uses the true P, and they note that it "does not lead directly to practical localization schemes" [27, Section 1]. Our competitors use the same oracle information, so the comparison is like for like.

1.3 The answer depends on the loss

We measure analysis error by a quadratic loss $(\hat{x}-x)^{\mathsf{T}}\Omega(\hat{x}-x)$ in physical variables. Two choices matter.

- *Physical squared error* ($\Omega = I$), as in Snyder and Hakim's experiments. Here the conjecture **fails at every finite ensemble size**, even locally: generically a nearby rotated basis does strictly better (Sections 3–5). It also fails at leading order in large ensembles whenever the criterion of Section 7 fails, which is the typical case.
- *Prior-normalised (Mahalanobis) squared error* ($\Omega = P^{-1}$), the natural loss that is invariant under changes of variables. Here the conjecture **holds at leading order** for every observation network, among complete localisations and unit-diagonal positive semidefinite tapers in whitened bases (Section 7). For a scalar observation at finite N, the canonical basis is a stationary point; it is a strict local minimum (up to the completion of unobserved directions) if a scalar $\zeta > 0$, and not a local minimum if $\zeta < 0$. The inequality $\zeta > 0$ holds in every case we computed but is not proved, and the case $\zeta = 0$ is unresolved (Section 8).

So the conjecture, as operationalised in Snyder and Hakim's experiments, is false; the loss-invariant version they emphasise survives at leading order.

1.4 Results at a glance

Table 1 lists the main results. "Canonical tapers" means deterministic symmetric Schur multipliers applied in canonical coordinates; "complete localisation" keeps only the sample variances in the chosen basis. Throughout, the prior covariance used to build any basis is the true one.

1.5 Relation to optimal-localisation and sampling-error theory

Optimal localisation in fixed coordinates. Flowerdew [7] derives the localisation factor that minimises expected analysis error for a single observation, of the form $Q^2/(1+Q^2)$ with Q the ratio of the true correlation to its sampling standard deviation [7, Section 2]; the factor vanishes when the true correlation vanishes. In canonical coordinates every relevant true correlation is zero, and our Lemma 3.1 is the exact finite-N form of that zero-correlation case. Flowerdew mostly neglects sampling error in variances [7, Section 2.2]; our Theorem 3.3 is driven precisely by it. Ménétrier and co-authors develop optimality criteria for linearly filtering sample covariances [18, 19] and for joint localisation and hybridisation [17]; Periáñez, Reich and Potthast [23] optimise localisation for ensemble Kalman filter systems; Anderson studies the need for localisation with a hierarchical ensemble filter [2] and corrects sampling error adaptively [3]; Morzfeld and Hodyss [20] show that many optimal schemes follow a common law, and Atkinson and co-authors [4] compare traditional and hybrid optimal localisation. All of these work in a fixed set of variables. Our question is the choice of variables itself. Remark 3.2 shows why covariance-error criteria cannot settle it: in whitened coordinates the covariance error of complete localisation is the same in every orthonormal basis.

Gain bias and sampling-error expansions. The positivity in Theorem 3.3 comes from the nonlinearity of the gain in the sample variance, the mechanism behind the gain bias of van Leeuwen [28] and Hodyss and Morzfeld [10]. Theorem 6.1 is a sampling-error expansion in the tradition of Sacher and Bartello [24], who express the analysis error in terms of the sampling noise in the background covariance, and of Furrer and Bengtsson [9], who study tapered estimators of

Table 1: Main results. "Proof" means a written proof in this paper (not machine-verified); "exact" means rational arithmetic in `exact_certificates.py`; "numerical" means floating-point evidence only.

Setting	Result	Status
Scalar observation, any $N \geq 2$, physical loss	A rotation plus complete localisation beats every canonical taper with unit observed diagonal whenever H^T is not an eigenvector of P (Thm 3.3, Cor. 3.4)	Proof
Same, with inflation	A rotation plus the *same* deterministic rescaling $\mathrm{diag}(\tau_0, 1, \ldots, 1)$ in the rotated basis beats every canonical taper with any observed-variance factor $\tau > 0$ (Thm 3.7); an unrescaled rotation need not (Remark 3.8)	Proof
Two-dimensional example	Explicit rational rotation, risk gap $\leq -1/20416536$ at $N = 3$ (Section 4.1, App. A)	Proof + exact
Aligned example, $N = 3$	Canonical basis is a strict local maximum (Prop. 4.1)	Proof + exact; other N numerical
Several observations, any N	Compared with rotated complete localisations only: not stationary if the loss couples an observed and an unobserved canonical direction (Cor. 5.2); stationary if $[W, C] = 0$ (Cor. 5.3)	Proof; converse conjectured
Several observations, large N	Uniform expansion with leading objective F (Thm 6.1); canonical minimises F iff commutation and opposite ordering (Thm 7.1); if not, beaten for all large N (Cor. 7.6)	Proof
Prior-normalised loss	Leading-order optimal among complete localisations and unit-diagonal tapers in whitened bases (Thm 7.1, Rem. 7.5); scalar observation, finite N: strict local minimum modulo completion if $\zeta_{\nu,a,\beta} > 0$, not a local minimum if $\zeta < 0$, $\zeta = 0$ unresolved (Prop. 8.1)	Proof; $\zeta > 0$ numerical
Snyder–Hakim testbed (reconstruction)	For $N = 8$ to 64, the best basis found removes 42–54% and ZCA 41–54% of canonical excess error; canonical best found under prior-normalised loss (Section 10)	Numerical

prior and posterior covariances in Kalman filter variants; we have not compared their expansions term by term with ours. The new element is uniformity over the basis, which makes $F(U)$ a basis-selection objective. Non-asymptotic bounds for ensemble updates with localisation are given by Al-Ghattas and Sanz-Alonso [1].

Localisation in transformed variables. Choosing a basis and then localising, often diagonally, is common practice: spectral and wavelet-diagonal correlation models and spectral localisation [6, 16], localisation in streamfunction–velocity potential variables to preserve balance [13], transformed localisation for non-local observation operators [22], and model-space localisation in serial filters [25]. Lorenc notes that the optimal settings depend on the error criterion [16]. Our criterion (Theorem 7.1) gives a precise version of that remark for complete localisation, and Corollary 10.1 applies it to Fourier-diagonal localisation.

Whitening and shrinkage in statistics. Kessy, Lewin and Strimmer [14] show that the choice among whitening transformations, which differ by a rotation, should depend on the purpose; symmetric (ZCA) whitening is one of their canonical choices and turns out to be nearly optimal in our testbed. Ledoit–Wolf shrinkage [15] keeps the sample eigenbasis and adjusts the eigenvalues; complete localisation in a fixed basis does the opposite. Finally, Snyder [26] uses canonical coordinates to analyse EnKF sampling error with random-matrix approximations; that abstract does not address the choice of localisation basis.

The sufficiency half of Theorem 7.1 follows from a known antimonotone trace inequality stated by

Mostajeran and Sepulchre [21], who attribute it to Bourin [5] and Fujii [8]; we do not claim it. Our contributions are the exact finite-N obstructions and examples, the uniform expansion as a basis objective with its necessary-and-sufficient criterion, the finite-N analysis under prior-normalised loss, and the application to Snyder and Hakim's test case. A literature search, bounded and dated 3 November 2026, found no earlier resolution of the conjecture; this is not a guarantee of priority.

1.6 Roadmap and notation

Section 2 sets up one analysis step and the exact conditional risk. Section 3 proves the finite-N obstruction for a scalar observation, including inflation. Section 4 gives the two explicit examples. Section 5 treats several observations at finite N. Sections 6–7 derive the large-N objective and the criterion. Section 8 studies prior-normalised loss at finite N. Section 9 gives a closed-form two-dimensional rotation and its size, and Section 10 applies everything to Snyder and Hakim's test case. Section 11 discusses implications and open problems. Table 2 collects the notation.

Table 2: Notation.

Symbol	Meaning
$x = Lz$, $P = LL^\mathsf{T}$	physical state; canonical (whitened) state $z \sim N(0, I_m)$
$W = L^\mathsf{T}L$	physical squared loss written in canonical coordinates; $W = I$ is prior-normalised loss
N, $\nu = N - 1$	ensemble size and degrees of freedom
a	$1 + 1/N$ (ensemble-mean update) or 1 (known prior mean)
S, D_i	sample covariance of z; sample variances, i.i.d. χ^2_ν/ν
$\beta = h^2$; β_i	scalar observation strength; canonical strengths, $G^\mathsf{T}G = \mathrm{diag}(\beta_i)$
T, $T \circ S$	deterministic taper and Schur (entrywise) product
Q_θ, U	plane rotation; orthonormal basis $U = [u_1, \dots, u_m]$
$\mathcal{R}$, $\mathcal{R}_N$	expected analysis risk at ensemble size N
$\gamma_{\nu,a,\beta}$, $\Phi_{\nu,a}$	first-variation constants (Thm 3.3, Prop. 5.1)
$\kappa_{\nu,a,\beta}$, $\omega_{\nu,\beta}$, $\zeta_{\nu,a,\beta}$	curvature constants (Prop. 4.1, Prop. 8.1)
C; $B = I - C$; $A = CWC$	posterior covariance; information matrix; loss-weighted posterior matrix (all in canonical coordinates)
$F(U)$	leading basis objective $\sum_i \mu_i(U)\lambda_i(U)$, with $\lambda_i(U) = u_i^\mathsf{T}Au_i$, $\mu_i(U) = u_i^\mathsf{T}Bu_i$
$\|\cdot\|$	spectral norm unless marked $\|\cdot\|_F$ (Frobenius)

2 One analysis with a Gaussian ensemble

Let $m \geq 2$, let $L \in \mathbb{R}^{m\times m}$ be invertible, and write the physical state and its prior covariance as

$$x = Lz, \qquad z \sim N(0, I_m), \qquad P = LL^\mathsf{T}, \qquad W = L^\mathsf{T}L \succ 0.$$

Physical squared error $|\hat{x} - x|^2$ equals $(\hat{z} - z)^\mathsf{T}W(\hat{z} - z)$; this is why W appears. Dividing by the dimension, as Snyder and Hakim do, changes no comparison.

Scalar observation. After whitening the observation noise (variance $R_\mathrm{o} > 0$) and choosing canonical coordinates, the observation is

$$y = hz_1 + \varepsilon, \qquad h > 0, \qquad \varepsilon \sim N(0, 1), \qquad \beta = h^2.$$

The truth, the noise and $N \geq 2$ ensemble members $z^{(r)} \sim N(0, I_m)$ are mutually independent.

Define

$$\bar{z} = \frac{1}{N}\sum_{r=1}^{N} z^{(r)}, \qquad S = \frac{1}{\nu}\sum_{r=1}^{N}(z^{(r)} - \bar{z})(z^{(r)} - \bar{z})^{\mathsf{T}}, \qquad \nu = N - 1, \qquad a = 1 + \frac{1}{N}.$$

Arrange the members as the columns of an $m \times N$ matrix and multiply on the right by an orthogonal $N \times N$ matrix whose first column is $N^{-1/2}\mathbf{1}$. The new entries are again independent standard normals; the first column is $\sqrt{N}\bar{z}$, and νS is the sum of the outer products of the remaining ν columns, the *contrast columns*. Hence $\bar{z}$ is independent of S, with covariance I_m/N, and

$$D_i := S_{ii} \text{ are independent}, \qquad D_i \sim \chi^2_\nu/\nu, \qquad \mathbb{E}D_i = 1, \qquad \operatorname{var}(D_i) = 2/\nu.$$

The same holds in any fixed orthonormal basis U: $U^{\mathsf{T}}SU$ has the law of S, so the diagonal entries of $U^{\mathsf{T}}SU$ are again independent χ^2_ν/ν variables, independent of the sample mean.

For a covariance estimate $M = M(S) \succeq 0$ the analysis mean is

$$\widehat{z}_M = \bar{z} + k(M)\,(y - h\bar{z}_1), \qquad k(M) = \frac{hMe_1}{1 + \beta M_{11}}, \tag{2.1}$$

and its physical risk is $\mathcal{R}(M) = \mathbb{E}[(\widehat{z}_M - z)^{\mathsf{T}}W(\widehat{z}_M - z)]$. Since $\widehat{z} - z = (I - hke_1^{\mathsf{T}})(\bar{z} - z) + k\varepsilon$ and $\bar{z} - z$ has covariance aI, conditioning on S gives the exact identities

$$\mathcal{E}(k) = a(I - hke_1^{\mathsf{T}})(I - hke_1^{\mathsf{T}})^{\mathsf{T}} + kk^{\mathsf{T}}, \tag{2.2}$$

$$r(k) := \operatorname{tr}(W\mathcal{E}(k)) = a\operatorname{tr}W - 2ah\,e_1^{\mathsf{T}}Wk + (a\beta + 1)\,k^{\mathsf{T}}Wk. \tag{2.3}$$

If the true prior mean replaces the ensemble mean, the same identities hold with $a = 1$, keeping the same centred estimator S. Every finite-N result below covers both conventions.

Complete localisation in a basis. For an orthonormal U, complete localisation keeps only the sample variances of $w = U^{\mathsf{T}}z$: $M_U = U\operatorname{diag}(U^{\mathsf{T}}SU)U^{\mathsf{T}}$. By the remark above, M_U has the law of $U\operatorname{diag}(D_1, \ldots, D_m)U^{\mathsf{T}}$. The update is formed with the observation row in the new coordinates and the analysis is mapped back.

Remark 2.1 (Rotating the observations changes nothing). For several whitened observations $y = Gz + \varepsilon$ and any orthogonal O acting on the observations, the gain $K = MG^{\mathsf{T}}(I + GMG^{\mathsf{T}})^{-1}$ becomes KO^{T} and the increment $K(y - G\bar{z})$ is unchanged. Hence, once the state covariance is localised, only the state basis matters; we rotate only the state.

3 Scalar observation: failure at every finite ensemble size

3.1 The best canonical taper

For a deterministic symmetric $T \in \mathbb{R}^{m\times m}$ with $T_{11} = 1$, set $M = T \circ S$. Positive semidefiniteness of T is not needed. Let $\mathcal{R}_{\text{can}}$ be the risk of complete canonical localisation, $T = I_m$.

Lemma 3.1 (Best canonical taper)**.** *With $D \sim \chi^2_\nu/\nu$,*

$$\mathcal{R}(T \circ S) - \mathcal{R}_{\text{can}} = \frac{\beta(a\beta + 1)}{\nu}\,\mathbb{E}\frac{D}{(1 + \beta D)^2}\sum_{j=2}^{m} W_{jj}T_{j1}^2 \;\geq 0, \tag{3.1}$$

with equality exactly when $T_{j1} = 0$ for all $j > 1$. Entries of T outside its first row and column do not affect the update.

Proof. The first gain component is $hD_1/(1+\beta D_1)$ and the others are $hT_{j1}S_{j1}/(1+\beta D_1)$. In the contrast representation of Section 2, condition on the first coordinate of every contrast column. Then D_1 is fixed and $(S_{21},\dots,S_{m1})$ is centred Gaussian with covariance $(D_1/\nu)I_{m-1}$. Hence all terms of (2.3) that are linear in an unobserved gain component, and all products of two different unobserved components, have zero conditional mean. Expanding (2.3) gives (3.1). The expectation is positive and $W_{jj}>0$ because $W \succ 0$. □

Remark 3.2 (Covariance-error criteria are blind to the basis). In whitened coordinates the true covariance is I in every orthonormal basis, and $\mathbb{E}\,\|M_U - I\|_F^2 = \sum_i \operatorname{var}(D_i) = 2m/\nu$ for every U. A criterion based only on the error of the localised covariance cannot prefer one whitened basis to another. The basis matters only through its effect on the gain and the analysis error, which is where W enters.

3.2 A strict first-variation obstruction

Let Q_θ be the identity outside the $(1,j)$-plane, with block $\left(\begin{smallmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{smallmatrix}\right)$ in that plane. Transform to $w = Q_\theta^{\mathsf T} z$, keep the sample variances in those coordinates, update with the observation row $he_1^{\mathsf T}Q_\theta$, and map the analysis back. Denote the physical risk of this procedure by $\mathcal{R}_N(\theta)$. By Section 2, for each fixed θ,

$$\mathcal{R}_N(\theta) = \mathbb{E}\, r(k(M_\theta)), \qquad M_\theta = Q_\theta \operatorname{diag}(D_1,\dots,D_m) Q_\theta^{\mathsf T}, \qquad D_i \text{ i.i.d. } \chi^2_\nu/\nu. \tag{3.2}$$

Holding the D_i fixed while θ varies is a coupling used to differentiate an expectation; it does not mean that one realised ensemble keeps its variances when rotated.

The mechanism in words. In canonical coordinates the gain has a single nonzero component, along the observed direction, and its size depends on the sampled variance D_1. Sampling error in D_1 makes this gain too large or too small, and no canonical taper can help, because it can only scale entries whose true value is zero. After a small rotation, the gain acquires a component along an unobserved direction j, proportional to $D_1 - D_j$. This cross component is correlated with the error in D_1: it is large precisely when the observed variance has been overestimated. When the loss couples the two directions ($W_{1j} \neq 0$), this correlation can be used to cancel part of the error caused by the misestimated gain. The size of the effect is the constant γ below.

Theorem 3.3 (Exact first variation)**.** *For every $N \geq 2$, $h>0$ and $j>1$,*

$$\mathcal{R}'_N(0) = 2\beta W_{1j}\,\gamma_{\nu,a,\beta}, \qquad \gamma_{\nu,a,\beta} := \mathbb{E}\frac{(D-1)(D-a)}{(1+\beta D)^2} > 0. \tag{3.3}$$

Consequently, if $W_{1j} \neq 0$, every sufficiently small rotation with sign opposite to W_{1j} strictly improves on every deterministic symmetric taper with $T_{11}=1$ in canonical coordinates.

Proof. At $\theta=0$, $(M_0)_{11} = D_1$ and $(M_0)_{j1}=0$, while $(M_\theta)'_{11}|_0 = 0$ and $(M_\theta)'_{j1}|_0 = D_1 - D_j$. Thus $k_1(0) = hD_1/(1+\beta D_1)$, $k_1'(0)=0$, $k_j(0)=0$ and $k_j'(0) = h(D_1-D_j)/(1+\beta D_1)$. Differentiating (2.3),

$$r'(0) = 2W_{1j}k_j'(0)\big[-ah + (a\beta+1)k_1(0)\big] = \frac{2\beta W_{1j}(D_1-D_j)(D_1-a)}{(1+\beta D_1)^2}.$$

Integrating out D_j ($\mathbb{E}D_j = 1$) gives (3.3). Differentiation under the expectation is valid: the denominator is at least one, and on a compact angle interval the θ-derivative of the conditional risk is bounded by a polynomial in $\sum_i D_i$, all of whose moments are finite.

For the sign, write $\gamma_{\nu,a,\beta} = \mathbb{E}[(D-1)^2(1+\beta D)^{-2}] - (a-1)\mathbb{E}[(D-1)(1+\beta D)^{-2}]$. The first term is positive. For the strictly decreasing $f(d) = (1+\beta d)^{-2}$ and an independent copy D' of D, $\mathbb{E}[(D-1)f(D)] = \operatorname{Cov}(D, f(D)) = \frac{1}{2}\mathbb{E}[(D-D')(f(D)-f(D'))] < 0$. Since $a \geq 1$, $\gamma_{\nu,a,\beta} > 0$. A nonzero derivative gives a strictly improving nearby angle, and Lemma 3.1 extends the comparison to all canonical tapers. □

Corollary 3.4 (Physical-coordinate condition)**.** *Let $P \succ 0$, let $H \neq 0$ be a scalar observation row and $R_o > 0$ its error variance. Under physical squared loss, if H^T is not an eigenvector of P, canonical localisation is not locally optimal at any ensemble size $N \geq 2$. The condition does not depend on how the unobserved canonical directions are completed.*

Proof. A canonical square root exists: take any L_0 with $L_0 L_0^\mathsf{T} = P$, let $v = L_0^\mathsf{T} H^\mathsf{T} / |L_0^\mathsf{T} H^\mathsf{T}|$, choose an orthogonal O with $Oe_1 = v$, and set $L = L_0 O$. Then $L^\mathsf{T} H^\mathsf{T} = |L_0^\mathsf{T} H^\mathsf{T}| e_1$, so $HL/\sqrt{R_o} = he_1^\mathsf{T}$. By Theorem 3.3, stationarity requires $We_1 \parallel e_1$ with $W = L^\mathsf{T} L$. Multiplying $We_1 = \lambda e_1$ by $L^{-\mathsf{T}}$ gives $Le_1 \parallel H^\mathsf{T}$, that is $PH^\mathsf{T} = L(L^\mathsf{T} H^\mathsf{T}) \parallel Le_1 \parallel H^\mathsf{T}$; the converse reverses the steps. The condition $PH^\mathsf{T} \parallel H^\mathsf{T}$ involves only P and H. □

This is an exact statement for every finite N. When the first derivative vanishes, nothing follows about global optimality; Section 4.2 shows that the canonical basis can then even be a local maximum.

3.3 Robustness to variance inflation

Practitioners rescale sampled variances (multiplicative inflation or deflation). We now let the canonical side do the same: allow every deterministic symmetric T with $T_{11} = \tau > 0$, with no condition on the other diagonal entries and no positive semidefiniteness. Put $X = \tau D$, $D \sim \chi^2_\nu / \nu$, and

$$\Psi_{\nu,a,\beta}(\tau) = \mathbb{E} \frac{a + \beta X^2}{(1 + \beta X)^2}, \qquad \gamma_{\nu,a,\beta}(\tau) = \mathbb{E} \frac{(X-1)(X-a)}{(1+\beta X)^2},$$

so that $\gamma_{\nu,a,\beta}(1) = \gamma_{\nu,a,\beta}$.

Lemma 3.5 (Inflated canonical tapers)**.** *For every such T,*

$$\mathcal{R}(T \circ S) = W_{11} \Psi_{\nu,a,\beta}(\tau) + a \sum_{j=2}^{m} W_{jj} + \frac{\beta(a\beta+1)}{\nu} \mathbb{E} \frac{D}{(1+\beta\tau D)^2} \sum_{j=2}^{m} W_{jj} T_{j1}^2. \tag{3.4}$$

Hence $\inf_T \mathcal{R}(T \circ S) = W_{11} \Psi^* + a \sum_{j \geq 2} W_{jj}$, with $\Psi^* = \inf_{\tau > 0} \Psi_{\nu,a,\beta}(\tau)$, and minimising sequences may be taken with $T_{j1} = 0$ for $j > 1$.

Proof. Write $g = (1 + \beta\tau D_1)^{-1}$. The gain is $k_1 = h\tau D_1 g$ and $k_j = h T_{j1} S_{j1} g$ ($j \geq 2$); entries of T outside its first row and column never enter. Expanding (2.3),

$$r = a \operatorname{tr} W + \big[-2ahW_{11}k_1 + (a\beta+1)W_{11}k_1^2\big] + \sum_{j \geq 2} \big[-2ahW_{1j} + 2(a\beta+1)W_{1j}k_1\big]k_j + (a\beta+1) \sum_{j,l \geq 2} W_{jl} k_j k_l.$$

As in Lemma 3.1, conditionally on the first coordinates of the contrast columns the S_{j1}, $j \geq 2$, are independent $N(0, D_1/\nu)$, so the middle sum and the off-diagonal part of the last sum have conditional mean zero, and $\mathbb{E}[k_j^2 \mid D_1] = \beta T_{j1}^2 D_1 g^2 / \nu$. Finally $a - 2ahk_1 + (a\beta+1)k_1^2 = (a + \beta\tau^2 D_1^2) g^2$, as one checks by expanding $(1+\beta\tau D_1)^2$. All terms are integrable since $|k_1| \leq 1/h$ and $|k_j| \leq h|T_{j1}||S_{j1}|$. □

Lemma 3.6 (The best inflation factor)**.** *Fix $\nu \geq 1$, $a \geq 1$, $\beta > 0$ and write $\Psi = \Psi_{\nu,a,\beta}$.*

1. *Ψ is continuously differentiable on $(0, \infty)$, with $\tau \Psi'(\tau) = 2\beta\, \mathbb{E}[X(X-a)(1+\beta X)^{-3}]$, $\Psi(0+) = a$, $\Psi'(0+) = -2a\beta$ and $\Psi(\infty) = 1/\beta$. The derivative Ψ' changes sign at most once, from negative to positive, so Ψ has at most one stationary point; a minimiser $\tau^* \in (0, \infty)$, if it exists, is that point.*
2. *If $N \geq 3$, a minimiser exists.*

3. *If $N = 2$, a minimiser exists if and only if $a\beta < 3$. If $a\beta \geq 3$, then $\Psi(\tau) > 1/\beta$ for every τ, and $\Psi^* = 1/\beta$ is approached only as $\tau \to \infty$, that is, by the deterministic gain $k = h^{-1}e_1$.*

Proof. (1) The integrand of Ψ is bounded by $\max(a, 1/\beta)$, because $a + \beta u^2 \leq \max(a, 1/\beta)(1 + \beta^2 u^2) \leq \max(a, 1/\beta)(1 + \beta u)^2$. Its τ-derivative $2\beta D(\tau D - a)(1 + \beta\tau D)^{-3}$ is dominated by $2\beta D(\tau_1 D + a)$ for $\tau \leq \tau_1$. Dominated convergence gives the formula and the limits; as $\tau \to \infty$ the integrand tends to $1/\beta$ because $D > 0$ almost surely.

For the sign pattern, let $\phi(x) = x(x - a)(1 + \beta x)^{-3}$, which is negative on $(0, a)$ and positive on (a, ∞), and let f_τ be the density of $X = \tau D$, proportional to $\tau^{-\nu/2} x^{\nu/2-1} e^{-\nu x/(2\tau)}$. For $\tau > \tau_0$ the ratio $\ell(x) = f_\tau(x)/f_{\tau_0}(x)$ is a positive constant times $\exp\{\frac{\nu}{2}(\tau_0^{-1} - \tau^{-1})x\}$, strictly increasing in x. Hence $\phi(x)\ell(x) \geq \phi(x)\ell(a)$ for every $x > 0$, strictly unless $x = a$: both sides are negative with $\ell(x) < \ell(a)$ when $x < a$, and positive with $\ell(x) > \ell(a)$ when $x > a$. Integrating against f_{τ_0},

$$\mathbb{E}\phi(\tau D) = \int \phi\, \ell\, f_{\tau_0} > \ell(a) \int \phi\, f_{\tau_0} = \ell(a)\, \mathbb{E}\phi(\tau_0 D).$$

So once $\Psi'(\tau_0) \geq 0$, $\Psi'(\tau) > 0$ for all $\tau > \tau_0$. This is the monotone-likelihood-ratio argument; it gives at most one sign change, from negative to positive.

Write $y = \beta X$ and $J(\tau) = \beta(\Psi(\tau) - 1/\beta) = \mathbb{E}[(a\beta - 1 - 2y)(1 + y)^{-2}]$. Since Ψ is continuous with end values a and $1/\beta$ and $\Psi'(0+) < 0$, a minimiser exists as soon as $J(\tau_0) < 0$ for some τ_0.

(2) The integrand of $\tau\mathbb{E}[y(1+y)^{-2}] = \mathbb{E}[\beta\tau^2 D(1+\beta\tau D)^{-2}]$ increases to $(\beta D)^{-1}$, so by monotone convergence the limit is $\mathbb{E}[(\beta D)^{-1}]$, which equals $\nu/(\beta(\nu - 2))$ for $\nu > 2$ and $+\infty$ for $\nu = 2$. Also $\tau(1 + \beta\tau D)^{-2} \leq (4\beta D)^{-1}$ and tends to zero, so for $\nu > 2$ dominated convergence gives $\tau\mathbb{E}(1 + y)^{-2} \to 0$; for $\nu = 2$, D is standard exponential and $\tau\mathbb{E}(1 + y)^{-2} \leq \tau \int_0^\infty (1 + \beta\tau d)^{-2}\, dd = 1/\beta$. In both cases $\tau J(\tau) \to -2\mathbb{E}[(\beta D)^{-1}] < 0$ or $-\infty$, so $J < 0$ for large τ.

(3) Let $\nu = 1$, so $D = Z^2$ with Z standard normal, and put $c = a\beta$ and $s = \beta\tau$, so $y = sZ^2$. *Existence for $c < 3$.* The density of y is $(2\pi s)^{-1/2} y^{-1/2} e^{-y/(2s)}$; substituting $y = \sigma^2$,

$$(2\pi s)^{1/2} J(\tau) = 2\int_0^\infty \frac{c - 1 - 2\sigma^2}{(1 + \sigma^2)^2}\, e^{-\sigma^2/(2s)}\, d\sigma \;\longrightarrow\; \int_0^\infty \frac{(c - 1 - 2y)\, y^{-1/2}}{(1 + y)^2}\, dy = \frac{\pi}{2}(c - 3)$$

as $s \to \infty$, by dominated convergence (the integrand is bounded by $\max(|c - 1|, 2)\min(1, \sigma^{-2})$) and two Beta integrals, $\mathrm{B}(\frac{1}{2}, \frac{3}{2}) = \mathrm{B}(\frac{3}{2}, \frac{1}{2}) = \pi/2$. If $c < 3$, $J < 0$ for large τ and a minimiser exists. *Non-existence for $c \geq 3$.* Stein's identity $\mathbb{E}[Zg(Z)] = \mathbb{E}[g'(Z)]$ with $g(z) = z\, u(z^2)$ gives $\mathbb{E}[(Z^2 - 1)u(Z^2)] = 2\mathbb{E}[Z^2 u'(Z^2)]$. Take $u(x) = (1 + sx)^{-1}$ and write $m_1 = \mathbb{E}(1 + y)^{-1}$, $m_2 = \mathbb{E}(1+y)^{-2}$. Using $\mathbb{E}[y(1+y)^{-2}] = m_1 - m_2$, the identity reads $(1 - m_1)/s - m_1 = -2(m_1 - m_2)$. Therefore, for $c = 3$,

$$J(\tau) = 2\,\mathbb{E}\frac{1 - y}{(1 + y)^2} = 2(2m_2 - m_1) = \frac{2(1 - m_1)}{s} > 0,$$

because $m_1 < 1$. For $c > 3$, J is larger still, since J increases with c. Hence $\Psi > 1/\beta$ everywhere, while $\Psi(\tau) \to 1/\beta$; the infimum is not attained. The limiting gain is $\lim_{\tau\to\infty} h\tau D_1(1 + \beta\tau D_1)^{-1} e_1 = h^{-1} e_1$. □

Theorem 3.7 (Inflation-robust obstruction)**.** *Let $N \geq 2$, $h > 0$, and $W_{1j} \neq 0$ for some $j > 1$. There are a rotation Q_θ in the $(1, j)$-plane and $\tau_0 > 0$ such that rotating by Q_θ, keeping the sample variances in the rotated coordinates, multiplying the first of them by τ_0 (taper $\mathrm{diag}(\tau_0, 1, \ldots, 1)$) and updating gives physical risk strictly below $\inf_T \mathcal{R}(T \circ S)$, the infimum over all deterministic symmetric T with $T_{11} > 0$. More precisely:*

1. *If a minimiser τ^* of Ψ exists (always for $N \geq 3$; for $N = 2$ iff $a\beta < 3$), take $\tau_0 = \tau^*$. The rotated risk $\mathcal{R}_N^{\tau^*}(\theta)$ satisfies*

$$\frac{d}{d\theta}\mathcal{R}_N^{\tau^*}(\theta)\Big|_{\theta=0} = 2\beta W_{1j}\, \gamma_{\nu,a,\beta}(\tau^*), \qquad \gamma_{\nu,a,\beta}(\tau^*) > 0, \tag{3.5}$$

and every sufficiently small θ with sign opposite to W_{1j} works.

2. *Otherwise ($N = 2$, $a\beta \geq 3$), the conclusion holds for every sufficiently small θ with sign opposite to W_{1j} and every sufficiently large τ_0, depending on θ.*

The positivity argument in Case 1 works at *any* stationary point of Ψ, so it does not depend on the uniqueness part of Lemma 3.6. The only dichotomy used is this: either a finite minimiser exists, and it is stationary, or the infimum is approached as $\tau \to \infty$.

Proof. Coupling and derivative. For fixed θ the diagonal of $Q_\theta^\mathsf{T} S Q_\theta$ has independent χ^2_ν/ν entries, independent of the sample mean. So $\mathcal{R}_N^\tau(\theta) = \mathbb{E}\, r(k(M_\theta))$, where

$$M_\theta = Q_\theta \operatorname{diag}(\tau D_1, D_2, \dots, D_m)\, Q_\theta^\mathsf{T}.$$

With $\cos\theta$ and $\sin\theta$ abbreviated c, s,

$$(M_\theta)_{11} = \mathrm{c}^2 \tau D_1 + \mathrm{s}^2 D_j, \qquad (M_\theta)_{j1} = \mathrm{cs}(\tau D_1 - D_j), \qquad (M_\theta)_{l1} = 0 \ (l \neq 1, j).$$

At $\theta = 0$, with $g = (1 + \beta\tau D_1)^{-1}$: $k_1 = h\tau D_1 g$, $k_1' = 0$, $k_j = 0$ and $k_j' = h(\tau D_1 - D_j)g$. Differentiating (2.3), $r'(0) = 2W_{1j}k_j'[-ah + (a\beta + 1)k_1] = 2\beta W_{1j}(\tau D_1 - D_j)(\tau D_1 - a)g^2$, and integrating out D_j gives (3.5). Differentiation under $\mathbb{E}$ is justified as in Theorem 3.3.

Sign at any stationary point. Let τ^* be stationary, $X = \tau^* D$, $g = (1+\beta X)^{-1}$ and $\psi = X(X-a)g^3$. By Lemma 3.6(1), $\mathbb{E}\psi = 0$. Put

$$\varrho(x) = \frac{(x-1)(1+\beta x)}{x} = \beta x + (1-\beta) - \frac{1}{x}, \qquad \varrho'(x) = \beta + x^{-2} > 0.$$

Then $(X-1)(X-a)g^2 = \psi\, \varrho(X)$, so $\gamma_{\nu,a,\beta}(\tau^*) = \mathbb{E}[\psi\varrho(X)] = \mathbb{E}[\psi\,(\varrho(X) - \varrho(a))]$. Both factors have the sign of $X - a$, so the integrand is nonnegative and vanishes only on the null event $X = a$. Hence $\gamma_{\nu,a,\beta}(\tau^*) > 0$. Integrability holds because $|\psi\varrho(X)| \leq |X-1||X-a|$ and $|\psi| \leq X|X-a|$.

Case 1. By Lemma 3.5, $\mathcal{R}_N^{\tau^*}(0) = W_{11}\Psi(\tau^*) + a\sum_{j\geq 2} W_{jj} = \inf_T \mathcal{R}(T \circ S)$. A nonzero derivative gives the claim.

Case 2. Now $\inf_T \mathcal{R}(T \circ S) = W_{11}/\beta + a\sum_{l\geq 2} W_{ll}$. Fix $|\theta| < \pi/4$ and put $t = \tan\theta$. As $\tau \to \infty$, $k \to h^{-1}(e_1 + te_j)$ on the event $D_1 > 0$, and the bounds $|k_1| \leq 1/h$, $|k_j| \leq |t|/h + h|\mathrm{cs}|D_j$ give dominated convergence. Hence $\mathcal{R}_N^\tau(\theta)$ tends to the risk of this deterministic gain,

$$\rho_\infty(t) = a \operatorname{tr} W - 2a(W_{11} + tW_{1j}) + \frac{a\beta + 1}{\beta}(W_{11} + 2tW_{1j} + t^2 W_{jj}),$$

with $\rho_\infty(0) = W_{11}/\beta + a\sum_{l\geq 2} W_{ll}$ and $\rho_\infty'(0) = 2W_{1j}/\beta \neq 0$. Choose a small t of sign opposite to W_{1j}, so that $\rho_\infty(t) < \rho_\infty(0)$, and then τ_0 so large that $\mathcal{R}_N^{\tau_0}(\theta) < \rho_\infty(0)$. □

Remark 3.8 (Scope and a numerical illustration). The competitor in Case 1 uses the *same* rescaling τ^* in the rotated basis as the best canonical taper; re-optimising it after rotation could only help. The matched rescaling is essential: an unrescaled rotation need not beat the best inflated canonical taper. For the example of Section 4.1 with $N = 3$ and the ensemble mean, the infimum over canonical tapers with any $T_{11} > 0$ is $2\Psi(\tau^*) + 4/3 = 2.7408$, while the best rotation with unrescaled complete localisation has risk 2.7453 (deterministic quadrature). Lemma 3.5 needs no positive semidefiniteness, but Theorem 3.7 still concerns deterministic tapers; sample-adaptive inflation is not covered. For the example of Section 4.1 with $N = 3$ and the ensemble mean, deterministic quadrature gives $\tau^* = 2.37206$ and $2\beta W_{12}\gamma(\tau^*) = 0.52893$. A seeded Monte Carlo of the actual ensemble procedure (4×10^6 ensembles, common random numbers, central difference with step 0.04) gives 0.5256 ± 0.0023, against 0.5260 for the exact central difference at that step. Over $N \in \{2, 3, 4, 5, 10, 20, 50\}$, both mean conventions and $\beta \in [0.05, 20]$, quadrature found $\gamma(\tau^*) > 0$ in every case, and a minimiser existed exactly when Lemma 3.6 predicts.

4 Two explicit counterexamples

4.1 A directly observed, correlated state

Take

$$L = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad H = (1, 0), \quad R_{\mathrm{o}} = 1, \quad W = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{4.1}$$

The first state variable is observed directly and is correlated with the second. Canonical coordinates are $z = (x_1, x_2 - x_1)$, the observation is $y = z_1 + \varepsilon$, and $\beta = 1$. Theorem 3.3 gives

$$\mathcal{R}'_N(0) = 2\,\mathbb{E}\frac{(D-1)(D-1-1/N)}{(1+D)^2} > 0,$$

so every sufficiently small negative rotation improves the risk, for every N. The competing physical transformation is $Q_\theta^{\mathsf{T}} L^{-1}$, followed by complete localisation.

Appendix A makes the angle explicit with rational entries for every N. For $N = 3$, put $t = 1/972216$ and

$$Q = \frac{1}{1+t^2}\begin{pmatrix} 1-t^2 & 2t \\ -2t & 1-t^2 \end{pmatrix}.$$

Then

$$\mathcal{R}_3(Q) - \mathcal{R}_3(I) \le -\frac{1}{20416536} < 0. \tag{4.2}$$

The bound is deliberately conservative: it certifies a strict improvement, and is not an estimate of the best improvement, which is of order 10^{-1} (Table 3).

4.2 Alignment can give a strict local maximum

For $m = 2$, $\beta > 0$ and diagonal loss $W = \mathrm{diag}(w_1, w_2)$, the risk is even in θ, so the first variation vanishes. Define

$$\kappa_{\nu,a,\beta} = \mathbb{E}\frac{(D-a)(D-1)}{(1+\beta D)^3}, \qquad \omega_{\nu,\beta} = \mathbb{E}\frac{(D-1)^2 + 2/\nu}{(1+\beta D)^2}.$$

Both are positive: for κ use the covariance argument of Theorem 3.3 with $f(d) = (1+\beta d)^{-3}$; ω is manifestly positive.

Proposition 4.1 (Exact curvature)**.** *In the diagonal-loss case,*

$$\mathcal{R}''_N(0) = -4\beta w_1 \kappa_{\nu,a,\beta} + 2\beta(a\beta+1) w_2\, \omega_{\nu,\beta}. \tag{4.3}$$

The canonical basis is a strict local maximum along this rotation family when $w_1/w_2 > (a\beta + 1)\omega_{\nu,\beta}/(2\kappa_{\nu,a,\beta})$, and a strict local minimum when the inequality is reversed.

Proof. Write $u = (M_\theta)_{11}$ and $v = (M_\theta)_{21}$, with independent $D_1, D_2 \sim \chi^2_\nu/\nu$. Then $u = D_1 + (D_2 - D_1)\theta^2 + O(\theta^4)$ and $v = (D_1 - D_2)\theta + O(\theta^3)$. The conditional risk is

$$w_1 \frac{a + \beta u^2}{(1+\beta u)^2} + w_2\Big(a + \frac{\beta(a\beta+1)v^2}{(1+\beta u)^2}\Big),$$

and the derivative of the first fraction in u is $2\beta(u-a)/(1+\beta u)^3$. Use $\mathbb{E}D_2 = 1$ and $\mathbb{E}[(D_1 - D_2)^2 \mid D_1] = (D_1 - 1)^2 + 2/\nu$. The polynomial domination of Theorem 3.3 justifies the second derivative under $\mathbb{E}$. □

Now take

$$P = \mathrm{diag}(4, 1), \qquad H = (1, 0), \qquad R_{\mathrm{o}} = 4, \qquad N = 3. \tag{4.4}$$

Here $z = (x_1/2, x_2)$; dividing the observation by two gives $\beta = 1$ and $W = \mathrm{diag}(4, 1)$. The observation direction is an eigenvector of P, so Corollary 3.4 does not apply. Since $\nu = 2$, D is exponential with mean one. Put $\eta = \int_0^\infty e^{-x}(1+x)^{-1}\,dx$. Integration by parts gives $\mathbb{E}(1+D)^{-2} = 1 - \eta$ and $\mathbb{E}(1+D)^{-3} = \eta/2$, whence

$$\kappa = (2a+5)\eta - a - 3, \qquad \omega = 6 - 9\eta, \qquad \mathcal{R}_3''(0) = 28a + 60 - (50a + 98)\eta.$$

An elementary exact lower bound on η suffices. With

$$p(x) = \frac{713 - 413x + 113x^2 - 13x^3 + x^4/2}{773},$$

nonnegativity of a square gives

$$0 \le \int_0^\infty (1+x)\Big(\frac{1}{1+x} - p(x)\Big)^2 e^{-x}\,dx = \eta - \frac{460}{773},$$

because $\int_0^\infty x^k e^{-x}\,dx = k!$ makes both $\int p e^{-x}$ and $\int (1+x)p^2 e^{-x}$ equal to $460/773$. Therefore

$$\mathcal{R}_3''(0) \le \begin{cases} -56/773, & a = 1 \quad \text{(known prior mean)}, \\ -508/773, & a = 4/3 \quad \text{(ensemble mean)}. \end{cases} \tag{4.5}$$

Both signs of small nonzero rotation improve the risk, and Lemma 3.1 extends the comparison to every canonical taper. Alignment of the observation with an eigenvector of P therefore does not rescue the conjecture. The example relies on a non-identity loss ($w_1/w_2 = 4$); with $W = I$ the same proposition gives a local minimum (Section 8).

Remark 4.2 (The aligned example at other ensemble sizes; numerical). Proposition 4.1 gives $\mathcal{R}_N''(0) = -16\kappa_{\nu,a,1} + 2(a+1)\omega_{\nu,1}$ for (4.4) at any N. Deterministic quadrature (not interval-certified) gives:

N	known mean	$N^2\times$	ensemble mean	$N^2\times$
2	$+0.52738$	$+2.11$	-0.71805	-2.87
3	-0.25941	-2.33	-0.86520	-7.79
5	-0.20207	-5.05	-0.40980	-10.25
10	-0.06772	-6.77	-0.11480	-11.48
100	-7.91×10^{-4}	-7.91	-1.20×10^{-3}	-11.99
1000	-7.99×10^{-6}	-7.99	-1.20×10^{-5}	-12.00

With the ensemble mean the curvature is negative at every tabulated N. With the known mean it is *positive* at $N = 2$ (a local minimum there) and negative for $3 \le N \le 1000$. The order-N^{-1} parts cancel, as they must, because the leading objective of Section 6 is constant here (Remark 7.3); the effect is of order N^{-2}, with $N^2\mathcal{R}_N''(0) \to -8$ and -12 numerically. The tabulated rows are a subset of the recorded grid $N \in \{2, 3, 4, 5, 10, 20, 50, 100, 1000\}$.

5 Several observations at finite ensemble size

Now let $y = Gz + \varepsilon$, $\varepsilon \sim N(0, I_k)$, in canonical coordinates: $G^{\mathsf{T}}G = \mathrm{diag}(\beta_1, \dots, \beta_m)$, with $\beta_i \ge 0$ and $\beta_i = 0$ for unobserved directions. Localisation is applied to the state sample covariance, and the observation-space covariance is formed as GMG^{T} from the localised M. For a scalar observation this coincides with Snyder and Hakim's rule of localising both $X(HX)^{\mathsf{T}}$ and

$(HX)(HX)^\mathsf{T}$; for several observations it is the convention we adopt. For complete localisation in the basis Q_θ, a rotation in the (i,j)-plane, the argument of Section 3 gives $\mathcal{R}_N(\theta) = \mathbb{E}\, r(K(M_\theta))$ with $M_\theta = Q_\theta \operatorname{diag}(D_1,\dots,D_m)Q_\theta^\mathsf{T}$, D_l i.i.d. χ^2_ν/ν,

$$K(M) = MG^\mathsf{T}(I_k + GMG^\mathsf{T})^{-1}, \qquad r(K) = a\,\mathrm{tr}(W(I-KG)(I-KG)^\mathsf{T}) + \mathrm{tr}(WKK^\mathsf{T}),$$

which follows from $\widehat{z} - z = (I-KG)(\bar{z}-z) + K\varepsilon$. Put $q_l = 1 + \beta_l D_l$.

Proposition 5.1 (Exact first variation, general observation matrix)**.** *For every $N \geq 2$, every G and every pair $i \neq j$,*

$$\mathcal{R}'_N(0) = 2W_{ij}\,\Phi_{\nu,a}(\beta_i,\beta_j), \qquad \Phi_{\nu,a}(\beta_i,\beta_j) = \mathbb{E}\Big[(D_i - D_j)\Big\{\frac{\beta_i(D_i-a)}{q_i^2 q_j} + \frac{\beta_j(D_j-a)}{q_i q_j^2}\Big\}\Big]. \quad (5.1)$$

Moreover $\Phi_{\nu,a}(x,y) = -\Phi_{\nu,a}(y,x)$, $\Phi_{\nu,a}(x,x) = 0$, and $\Phi_{\nu,a}(x,0) = x\,\gamma_{\nu,a,x} > 0$ for $x > 0$.

Proof. Let $\Lambda = \operatorname{diag}(D_l)$ and $V = I_k + G\Lambda G^\mathsf{T} \succeq I_k$. The push-through identity $G^\mathsf{T}(I_k + G\Lambda G^\mathsf{T})^{-1} = (I_m + G^\mathsf{T}G\Lambda)^{-1}G^\mathsf{T}$ and $G^\mathsf{T}G\Lambda = \operatorname{diag}(\beta_l D_l)$ give

$$G^\mathsf{T}V^{-1} = \operatorname{diag}(q_l^{-1})G^\mathsf{T}, \qquad K_0 := K(\Lambda) = \operatorname{diag}(D_l/q_l)G^\mathsf{T}, \qquad I - K_0 G = \operatorname{diag}(q_l^{-1}).$$

Derivative of the risk in K. With W symmetric, $dr = 2\,\mathrm{tr}\big(dK^\mathsf{T}W[K(I_k + aGG^\mathsf{T}) - aG^\mathsf{T}]\big)$. At K_0, $K_0(I + aGG^\mathsf{T}) - aG^\mathsf{T} = \operatorname{diag}((D_l - a)/q_l)G^\mathsf{T} =: \mathcal{D}_e G^\mathsf{T}$, using $G^\mathsf{T}GG^\mathsf{T} = \operatorname{diag}(\beta_l)G^\mathsf{T}$ and $D_l + a\beta_l D_l - a(1+\beta_l D_l) = D_l - a$.

Derivative of K in M. Differentiating $K = MG^\mathsf{T}V(M)^{-1}$ gives $dK = (I-KG)\,\dot{M}\,G^\mathsf{T}V^{-1}$, which at Λ is $\operatorname{diag}(q_l^{-1})\dot{M}\operatorname{diag}(q_l^{-1})G^\mathsf{T}$.

Rotation. At $\theta = 0$, $\dot{M} = \dot{Q}\Lambda + \Lambda\dot{Q}^\mathsf{T}$ with $\dot{Q} = e_j e_i^\mathsf{T} - e_i e_j^\mathsf{T}$, so $\dot{M}_{ij} = \dot{M}_{ji} = D_i - D_j$ and all other entries vanish (the convention of Theorem 3.3). Combining,

$$r'(0) = 2\sum_{l,p}\frac{\dot{M}_{lp}}{q_l q_p}W_{pl}\frac{D_l - a}{q_l}\beta_l = 2W_{ij}\frac{D_i - D_j}{q_i q_j}\Big[\frac{\beta_i(D_i-a)}{q_i} + \frac{\beta_j(D_j-a)}{q_j}\Big].$$

For differentiation under $\mathbb{E}$: $M_\theta \succeq 0$ gives $V(M_\theta) \succeq I$, hence $\|K\| \leq \|M_\theta\|\,\|G\| \leq \|G\|\sum_l D_l$, and $\|\partial_\theta K\|$ is bounded by a polynomial in $\sum_l D_l$. Taking expectations gives (5.1).

Antisymmetry follows by exchanging the i.i.d. D_i and D_j, and gives $\Phi_{\nu,a}(x,x) = 0$. If $\beta_j = 0$, then $q_j = 1$ and integrating out D_j gives $\beta_i\mathbb{E}[(D_i-1)(D_i-a)q_i^{-2}] = \beta_i\gamma_{\nu,a,\beta_i} > 0$. □

For $k = 1$ and $G = he_1^\mathsf{T}$ the proposition reduces to Theorem 3.3.

Corollary 5.2 (Finite-N obstruction for any observation matrix)**.** *Let $N \geq 2$ and G be arbitrary. If $W_{ij} \neq 0$ in canonical coordinates for an observed direction i ($\beta_i > 0$) and an unobserved direction j ($\beta_j = 0$), then a sufficiently small (i,j) rotation with sign opposite to W_{ij} strictly lowers the risk of canonical complete localisation. Canonical complete localisation is then not a local minimiser over rotated complete localisations, at any finite ensemble size.*

Proof. Proposition 5.1 gives $\mathcal{R}'_N(0) = 2\beta_i W_{ij}\gamma_{\nu,a,\beta_i} \neq 0$. □

The comparison here is with rotated complete localisations; we have not proved a finite-N analogue of Lemma 3.1 for several observations.

Corollary 5.3 (Commuting case is stationary)**.** *If $[W,C] = 0$, with $C = (I + G^\mathsf{T}G)^{-1}$, then canonical complete localisation is a stationary point of $\mathcal{R}_N$ on the orthogonal group for every $N \geq 2$, for every choice of canonical basis within the eigenspaces of $G^\mathsf{T}G$.*

Proof. In canonical coordinates $[W,C]_{ij} = W_{ij}\big((1+\beta_j)^{-1} - (1+\beta_i)^{-1}\big)$, so $[W,C]=0$ means $W_{ij}=0$ whenever $\beta_i \neq \beta_j$, while $\Phi_{\nu,a}(\beta_i,\beta_j)=0$ whenever $\beta_i=\beta_j$. The gradient at the identity is determined by the plane derivatives, and each vanishes by (5.1). □

Conjecture 5.4 (Sign of Φ; numerically supported)**.** $\Phi_{\nu,a}(x,y) > 0$ *for all* $x > y > 0$, $\nu \geq 1$ *and* $a \in \{1, 1+1/N\}$.

Deterministic quadrature over $\nu \in \{1,2,3,5,10,30\}$, both mean conventions and 55 ordered pairs $x > y$ drawn from $[0.003, 100]$ gave $\Phi > 0$ in all 660 cases; the smallest value of $\Phi/(x-y)$ was 1.7×10^{-8}, at $\nu = 30$, $(x,y) = (100, 30)$, where every term is small. Pointwise positivity of the integrand fails after symmetrisation, so a proof needs a different idea. If the conjecture holds, Corollary 5.3 becomes an equivalence: canonical complete localisation is stationary at finite N if and only if $[W,C]=0$, the same commutation condition as in Theorem 7.1.

6 A uniform large-ensemble expansion

Keep m, k, G, W fixed as $N \to \infty$, and define

$$C = (I_m + G^\mathsf{T} G)^{-1}, \qquad B = I_m - C = G^\mathsf{T}(I_k + GG^\mathsf{T})^{-1}G, \qquad A = CWC.$$

C is the Kalman posterior covariance in canonical coordinates, B measures the information supplied by the observations, and A weights posterior errors by the loss. For $U = [u_1, \ldots, u_m]$ orthogonal, complete localisation in basis U gives $M_U = U\,\mathrm{diag}(D_1, \ldots, D_m)U^\mathsf{T}$ in law and $K_U = K(M_U)$.

Theorem 6.1 (Uniform risk expansion)**.** *For the ensemble-mean update,*

$$\mathcal{R}_N(U) = \mathrm{tr}(WC) + \frac{\mathrm{tr}(WC^2)}{N} + \frac{2F(U)}{N-1} + O\big((N-1)^{-3/2}\big), \qquad F(U) = \sum_{i=1}^{m} (u_i^\mathsf{T} B u_i)(u_i^\mathsf{T} A u_i), \tag{6.1}$$

uniformly over $U \in O(m)$*; the implied constant depends only on* m, k, G, W*. For the known-mean update the term* $\mathrm{tr}(WC^2)/N$ *is absent.*

The first term is the Bayes risk. The sample-mean term does not depend on the basis at this order, so F is the leading objective for choosing a complete-localisation basis. We write $\lambda_i(U) = u_i^\mathsf{T} A u_i$ (loss weight) and $\mu_i(U) = u_i^\mathsf{T} B u_i$ (information weight), so $F = \sum_i \mu_i \lambda_i$. Each sample variance has relative error $\sqrt{2/\nu}$, and F prices that error in direction i at $\lambda_i \mu_i$.

The proof is a remainder bound that holds for a whole class of estimators. Let $\mathfrak{M}$ be the class of measurable maps $S \mapsto M(S)$ into symmetric matrices with

$$\text{(i) } M(S) \succeq 0, \qquad \text{(ii) } \mathbb{E}M(S) = I_m, \qquad \text{(iii) } \|M(S) - I_m\| \leq \|S - I_m\| \text{ almost surely,} \tag{6.2}$$

and write $E = M - I$. Let $V_0 = I_k + GG^\mathsf{T}$ and $K_0 = G^\mathsf{T} V_0^{-1}$.

Lemma 6.2 (Membership)**.** $\mathfrak{M}$ *contains (a) complete localisation* $M_U = U\,\mathrm{diag}(U^\mathsf{T} S U)U^\mathsf{T}$ *in any orthonormal basis* U*, and (b) every taper in a canonical basis* U_c*,* $M = U_c(T \circ (U_c^\mathsf{T} S U_c))U_c^\mathsf{T}$ *with* $T \succeq 0$ *and* $T_{ll} = 1$*.*

Proof. (a) $M_U \succeq 0$, $\mathbb{E}(U^\mathsf{T} S U)_{ll} = 1$, and $\|M_U - I\| = \max_l |u_l^\mathsf{T}(S-I)u_l| \leq \|S - I\|$. (b) $M \succeq 0$ by the Schur product theorem, and $\mathbb{E}M = U_c(T \circ I)U_c^\mathsf{T} = I$. For the norm, we use the classical bound $\|T \circ Y\| \leq \max_l T_{ll}\,\|Y\|$ for $T \succeq 0$ (see [11] for Schur multipliers); here is a short proof.

Write $T_{ij} = r_i^{\mathsf{T}} r_j$ with vectors $r_i \in \mathbb{R}^m$, and let $r^{(p)}$ be the vector of p-th components $(r_{1p}, \dots, r_{mp})$. For vectors x, y,

$$x^{\mathsf{T}}(T \circ Y)y = \sum_p (x \circ r^{(p)})^{\mathsf{T}} Y (y \circ r^{(p)}) \le \|Y\| \Big(\sum_p |x \circ r^{(p)}|^2\Big)^{1/2} \Big(\sum_p |y \circ r^{(p)}|^2\Big)^{1/2},$$

by Cauchy–Schwarz, and $\sum_p |x \circ r^{(p)}|^2 = \sum_i x_i^2 T_{ii} \le \max_l T_{ll} |x|^2$. Applied to $Y = U_c^{\mathsf{T}} S U_c - I$, this gives $\|M - I\| \le \|S - I\|$. □

Lemma 6.3 (Remainder)**.** *There is $\varkappa = \varkappa(m, k, G, W)$ such that for all $N \ge 2$ and all $M \in \mathfrak{M}$, the ensemble-mean update satisfies*

$$\Big|\mathcal{R}_N(M) - \operatorname{tr}(WC) - \frac{\operatorname{tr}(WC^2)}{N} - \mathbb{E}\operatorname{tr}(AEBE)\Big| \le \varkappa \nu^{-3/2}. \tag{6.3}$$

The known-mean update satisfies the same bound without $\operatorname{tr}(WC^2)/N$. We do not claim that the rate $\nu^{-3/2}$ is sharp (Remark 6.4).

Proof. Step 1: exact decomposition. For any K, the known-mean error covariance is $(I-KG)(I-KG)^{\mathsf{T}} + KK^{\mathsf{T}} = I - KG - G^{\mathsf{T}}K^{\mathsf{T}} + KV_0K^{\mathsf{T}}$. Since $K_0V_0 = G^{\mathsf{T}}$ and $G^{\mathsf{T}}V_0^{-1}G = B$, this equals $C + \Delta V_0 \Delta^{\mathsf{T}}$ with $\Delta = K - K_0$. The independent sample mean adds $N^{-1}(I-KG)(I-KG)^{\mathsf{T}}$. Hence, exactly,

$$\mathcal{R}_N(M) = \operatorname{tr}(WC) + \mathbb{E}\operatorname{tr}(W\Delta V_0 \Delta^{\mathsf{T}}) + \frac{1}{N}\mathbb{E}\operatorname{tr}(W(I-KG)(I-KG)^{\mathsf{T}}), \tag{6.4}$$

with no term linear in Δ.

Step 2: gain derivatives. For $M \succeq 0$, $V(M) = I + GMG^{\mathsf{T}} \succeq I$, so $\|V(M)^{-1}\| \le 1$ and $\|K(M)\| \le \|M\|\,\|G\|$. Along $M_t = I + tE$ with $\|E\| \le \frac{1}{2}$ we have $M_t \succeq \frac{1}{2}I$, $\|M_t\| \le \frac{3}{2}$, and, with $V_t = V(M_t)$,

$$\frac{d^2}{dt^2}K(M_t) = -2EG^{\mathsf{T}}V_t^{-1}GEG^{\mathsf{T}}V_t^{-1} + 2M_tG^{\mathsf{T}}V_t^{-1}GEG^{\mathsf{T}}V_t^{-1}GEG^{\mathsf{T}}V_t^{-1},$$

so $\big\|\frac{d^2}{dt^2}K(M_t)\big\| \le 2\|G\|^3(1 + \frac{3}{2}\|G\|^2)\|E\|^2$. The first derivative at $t = 0$ is $K_1(E) = (I - K_0G)EG^{\mathsf{T}}V_0^{-1} = CEG^{\mathsf{T}}V_0^{-1}$, with $\|K_1(E)\| \le \|G\|\,\|E\|$. On $\mathcal{G} = \{\|S - I\| \le \frac{1}{2}\}$, Taylor's formula with integral remainder gives $\Delta = K_1(E) + \Delta_2$ with $\|\Delta_2\| \le \varkappa_2 \|E\|^2$, $\varkappa_2 = \|G\|^3(1 + \frac{3}{2}\|G\|^2)$.

Step 3: moments and tail. Write $\nu S = \sum_{r=1}^{\nu} x_r x_r^{\mathsf{T}}$ with independent $x_r \sim N(0, I_m)$. Every entry of $S - I$ is an average of ν independent centred variables with finite moments of all orders, so $\mathbb{E}\|S - I\|^p \le \mathbb{E}\|S - I\|_F^p \le c_p \nu^{-p/2}$ (Marcinkiewicz–Zygmund) and $\sup_\nu \mathbb{E}\|S\|^p < \infty$. These variables have finite moment-generating functions near zero, so Cramér–Chernoff and $\|Y\| \le m \max_{ab}|Y_{ab}|$ give $\Pr(\mathcal{G}^c) \le 2m^2 e^{-c_0\nu}$ for some $c_0 > 0$.

Step 4: the quadratic term. As V_0 is symmetric and $G^{\mathsf{T}}V_0^{-1}V_0V_0^{-1}G = B$, $\operatorname{tr}(WK_1V_0K_1^{\mathsf{T}}) = \operatorname{tr}(WCEBEC) = \operatorname{tr}(AEBE)$. On $\mathcal{G}$, $\|E\| \le \frac{1}{2}$ by (iii) and

$$\big|\operatorname{tr}(W\Delta V_0\Delta^{\mathsf{T}}) - \operatorname{tr}(AEBE)\big| \le m\|W\|\,\|V_0\|\,(2\|G\|\,\varkappa_2\|E\|^3 + \varkappa_2^2\|E\|^4) \le \varkappa_3\|E\|^3.$$

On $\mathcal{G}^c$ both traces are bounded by a polynomial $\mathcal{P}(\|S\|)$, by (iii) and $\|K\| \le \|M\|\,\|G\|$; by Cauchy–Schwarz their contribution is at most $(\mathbb{E}\mathcal{P}^2)^{1/2}\Pr(\mathcal{G}^c)^{1/2} \le \varkappa_4 e^{-c_0\nu/2}$. Since $\mathbb{E}\|E\|^3 \le \mathbb{E}\|S - I\|^3 \le c_3\nu^{-3/2}$, we get $|\mathbb{E}\operatorname{tr}(W\Delta V_0\Delta^{\mathsf{T}}) - \mathbb{E}\operatorname{tr}(AEBE)| \le \varkappa_5\nu^{-3/2}$.

Step 5: the sample-mean term. Write $I - KG = C - \Delta G$. Then $\operatorname{tr}(W(I-KG)(I-KG)^{\mathsf{T}}) = \operatorname{tr}(WC^2) - 2\operatorname{tr}(W\Delta GC) + \operatorname{tr}(W\Delta GG^{\mathsf{T}}\Delta^{\mathsf{T}})$. By (ii) and linearity, $\mathbb{E}K_1(E) = C(\mathbb{E}E)G^{\mathsf{T}}V_0^{-1} = 0$,

so $|\mathbb{E}\,\mathrm{tr}(W\Delta GC)| \le m\,\|W\|\,\|G\|\,\mathbb{E}\,\|\Delta_2\|$, which is at most $\varkappa_6(\mathbb{E}\,\|E\|^2 + e^{-c_0\nu/2}) \le \varkappa_7/\nu$ after the same tail split. The quadratic term is at most $\varkappa_8\mathbb{E}\,\|\Delta\|^2 \le \varkappa_9/\nu$. So the last term of (6.4) is $\mathrm{tr}(WC^2)/N + O(N^{-1}\nu^{-1})$.

The constants depend only on m, k, G, W, through $\|G\|$, $\|W\|$, $\|V_0\|$ and the moment and tail constants of S; by (iii) none depends on the particular $M \in \mathfrak{M}$. Steps 4 and 5 give (6.3); dropping the sample-mean term gives the known-mean statement. □

Proof of Theorem 6.1. For M_U, $E_U = \sum_l (D_l - 1)u_l u_l^\mathsf{T}$ and, by independence and $\mathrm{var}(D_l) = 2/\nu$, $\mathbb{E}\,\mathrm{tr}(AE_UBE_U) = \frac{2}{\nu}\sum_l (u_l^\mathsf{T}Au_l)(u_l^\mathsf{T}Bu_l) = 2F(U)/\nu$. Lemmas 6.2 and 6.3 give (6.1), uniformly over U, with sample-mean error $O(N^{-1}\nu^{-1})$. □

Remark 6.4 (A sharper rate; sketch). For both classes in Lemma 6.2, E is a linear function of $S - I$. Carrying the expansion of K one order further gives $\Delta = K_1(E) + K_2(E, E) + O(\|E\|^3)$. The extra term $2\,\mathbb{E}\,\mathrm{tr}(WK_1V_0K_2^\mathsf{T})$ is a fixed linear combination of third moments of entries of $S - I$, which are $O(\nu^{-2})$ because third cumulants of averages of ν independent terms scale as ν^{-2}; also $\mathbb{E}\,\|E\|^4 = O(\nu^{-2})$. The remainder in (6.3) should therefore be $O(\nu^{-2})$, in line with numerical evidence. We record this only as a sketch.

Remark 6.5 (Fixed dimension versus practice). The expansion holds for fixed m and k as $N \to \infty$. Operational problems have $m \gg N$. Outside the fixed-dimension regime we use F as a basis-ranking heuristic, not as a risk prediction. In the 200-variable testbed of Section 10 the leading-order formula understates the canonical risk at $N = 4$, 8 and 16 and is accurate for it from $N = 32$, but it ranks the bases correctly at every N tested.

7 Exactly when the canonical basis minimises the leading objective

Canonical bases diagonalise B, equivalently $G^\mathsf{T}G$. Write $[A, B] = AB - BA$.

Theorem 7.1 (Sharp leading-order criterion)**.** *A canonical basis minimises F over $O(m)$ if and only if $[A, B] = 0$ and, in a simultaneous orthonormal eigenbasis with eigenvalues λ_i of A and μ_i of B,*

$$(\lambda_i - \lambda_j)(\mu_i - \mu_j) \le 0 \qquad \text{for all } i, j. \tag{7.1}$$

Repeated eigenvalues of B impose no ordering within their eigenspace. Every canonical basis has the same value $F = \mathrm{tr}(AB)$.

Sufficiency also follows from the antimonotone-pair inequality $\mathrm{tr}(AXBX) \ge \mathrm{tr}(ABX^2)$ [5, 8, 21] with $X = u_i u_i^\mathsf{T}$, summed over i. We give the elementary argument because it also yields the necessary conditions.

Proof. Work in any basis with $B = \mathrm{diag}(\mu_1, \dots, \mu_m)$. An (i, j)-plane rotation gives $F'(0) = 2(\mu_i - \mu_j)A_{ij}$. A minimiser is stationary, so $A_{ij} = 0$ whenever $\mu_i \ne \mu_j$; this is $[A, B] = 0$. Diagonalise A within the eigenspaces of B; this changes neither B nor the canonical value $\mathrm{tr}(AB)$. If a pair violates (7.1), a 45° mixing of that pair lowers F by $(\lambda_i - \lambda_j)(\mu_i - \mu_j)/2 > 0$. This proves necessity.

For sufficiency, use a simultaneous eigenbasis and let $M_{ij} = U_{ij}^2$; its rows and columns sum to one. Direct expansion gives

$$F(U) - \mathrm{tr}(AB) = -\frac{1}{2}\sum_j \sum_{i,l} M_{ij}M_{lj}(\lambda_i - \lambda_l)(\mu_i - \mu_l) \ge 0 \tag{7.2}$$

by (7.1). This also covers ties. □

Identity (7.2) has a simple reading. The matrix M is orthostochastic, and by the Schur–Horn theorem the diagonal of $U^{\mathsf{T}}BU$ is a mixture of the eigenvalues of B. Canonical bases sit at the extreme where information is concentrated, one eigenvalue per direction. When loss weight and information are oppositely ordered this extreme is best; when they are similarly ordered, mixing directions spreads the information and lowers F.

Since C is invertible and $B = I - C$, the commutation condition is equivalent to $[W, C] = 0$. With simultaneous eigenvalues c_i of C and w_i of W, we have $\lambda_i = c_i^2 w_i$ and $\mu_i = 1 - c_i$, so the ordering condition reads $(c_i^2 w_i - c_j^2 w_j)(c_i - c_j) \geq 0$. For $W = I$ this always holds. Therefore canonical coordinates minimise the leading objective under prior-normalised squared error, which in physical variables is the Mahalanobis loss with weight P^{-1}. This is the leading-order version of Snyder and Hakim's conjecture, and it holds for every observation network. The comparison class is complete localisation in orthonormal whitened bases and, by Remark 7.5, unit-diagonal positive semidefinite tapers in such bases. Inflation and non-orthogonal transformations are not covered; the latter give $\mathbb{E}M \neq I$ in whitened coordinates and hence an $O(1)$ bias. Under physical loss the commutation condition fails in the testbed of Section 10, and even when it holds the ordering can fail (Corollary 10.1). Physical loss does not by itself rule canonical coordinates out, however, as the next remark shows.

Remark 7.2 (Canonical coordinates can be optimal under physical loss). Take $P = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $H = I$ and $R_{\mathrm{o}} = I$, with physical loss. The canonical directions are the eigenvectors of P, with prior variances 2 and 4, so $W = \mathrm{diag}(2, 4)$, $C = \mathrm{diag}(1/3, 1/5)$, and $(\lambda_i, \mu_i) = (2/9, 2/3)$ and $(4/25, 4/5)$. Here $[A, B] = 0$ and $(\lambda_1 - \lambda_2)(\mu_1 - \mu_2) = -28/3375 < 0$, so by Theorem 7.1 the canonical basis minimises F. (Both mode variances exceed the observation-error variance, so Corollary 10.1 does not apply.) Whether canonical coordinates are leading-order optimal is decided by Theorem 7.1, not by the choice of loss alone.

Remark 7.3 (Ties). For the aligned example (4.4), $C = \mathrm{diag}(1/2, 1)$, $W = \mathrm{diag}(4, 1)$ and $A = I_2$, so F is constant over all bases. Yet the exact $N = 3$ curvature is negative (Proposition 4.1). When the leading objective ties, higher-order terms decide the finite-ensemble comparison.

7.1 Comparison with every canonical taper at large N

Theorem 7.1 compares complete localisations. The next lemma is the leading-order analogue of Lemma 3.1 for several observations.

Lemma 7.4 (Leading-order taper comparison)**.** *Let U_c be a canonical basis, so $U_c^{\mathsf{T}} B U_c = \mathrm{diag}(\mu_l)$ with $\mu_l = \beta_l/(1 + \beta_l)$, and let $A' = U_c^{\mathsf{T}} A U_c$. For a taper $T \succeq 0$ with $T_{ll} = 1$ in that basis (Lemma 6.2(b)),*

$$\mathbb{E}\,\mathrm{tr}(AEBE) = \frac{1}{\nu} \sum_l A'_{ll} \Big(2\mu_l + \sum_{p \neq l} \mu_p T_{lp}^2 \Big) \;\geq\; \frac{2}{\nu}\,\mathrm{tr}(AB) = \frac{2}{\nu} F(U_c),$$

with equality if and only if $A'_{ll} \mu_p T_{lp}^2 = 0$ for all $p \neq l$.

Proof. Put $Y = U_c^{\mathsf{T}} S U_c - I$, which has the law of $S - I$. Then

$$\mathrm{tr}(AEBE) = \sum_{l,s,p} A'_{ls} T_{sp} Y_{sp}\, \mu_p\, T_{pl} Y_{pl}.$$

For the Wishart matrix, $\mathrm{Cov}(S_{ab}, S_{cd}) = \nu^{-1}(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$, so $\mathbb{E}[Y_{sp} Y_{pl}] = \nu^{-1}(\delta_{sp}\delta_{pl} + \delta_{sl})$. Substituting gives the displayed expression since $T_{ll} = 1$. The inequality holds because $A'_{ll} \geq 0$ (as $A \succeq 0$) and $\mu_p \geq 0$, and $\sum_l A'_{ll} \mu_l = \mathrm{tr}(AB)$. □

Remark 7.5 (Tapers in any whitened basis). The same Wishart computation applies to a unit-diagonal taper $T \succeq 0$ in *any* orthonormal basis U (which lies in $\mathfrak{M}$ by the proof of Lemma 6.2(b)). With $A' = U^\mathsf{T} A U$ and $B' = U^\mathsf{T} B U$,

$$\nu\, \mathbb{E}\,\mathrm{tr}(AEBE) = \sum_{l,s} T_{ls}^2 (A'_{ls}B'_{ls} + A'_{ll}B'_{ss}) \;\geq\; 2F(U).$$

The terms with $l = s$ give $2F(U)$. For $l \neq s$, pair (l, s) with (s, l): since A' and B' are positive semidefinite, $|A'_{ls}B'_{ls}| \leq (A'_{ll}A'_{ss}B'_{ll}B'_{ss})^{1/2}$, and the arithmetic–geometric mean inequality gives $2A'_{ls}B'_{ls} + A'_{ll}B'_{ss} + A'_{ss}B'_{ll} \geq 0$. Hence, at leading order, partial tapering in a basis never beats complete localisation in the same basis, and when the criterion of Theorem 7.1 holds (in particular for $W = I$) no such estimator beats canonical complete localisation.

Corollary 7.6 (Large-N failure against every canonical taper)**.** *If a canonical basis does not minimise F, there are $U \in O(m)$, $\delta > 0$ and N_0 such that, for all $N \geq N_0$ and both mean conventions,*

$$\mathcal{R}_N(M_U) \leq \inf\{\mathcal{R}_N(M) : M \text{ a unit-diagonal PSD taper in some canonical basis}\} - \frac{\delta}{N-1}.$$

Proof. Choose U with $F(U) = \mathrm{tr}(AB) - \delta$, $\delta > 0$. By Lemmas 6.3 and 7.4, uniformly over canonical bases and tapers, $\mathcal{R}_N(M) \geq \mathrm{tr}(WC) + \epsilon_N + 2\,\mathrm{tr}(AB)/\nu - \varkappa\nu^{-3/2}$, where $\epsilon_N = \mathrm{tr}(WC^2)/N$ or 0. Also $\mathcal{R}_N(M_U) \leq \mathrm{tr}(WC) + \epsilon_N + 2(\mathrm{tr}(AB) - \delta)/\nu + \varkappa\nu^{-3/2}$. The difference is at most $-2\delta/\nu + 2\varkappa\nu^{-3/2} \leq -\delta/\nu$ once $\nu \geq (2\varkappa/\delta)^2$. □

Remark 7.7 (Scope of Corollary 7.6). The comparison class is narrower than in Lemma 3.1. The tapers must be positive semidefinite, so that $I + GMG^\mathsf{T}$ is invertible for every sample, and unit-diagonal, so that $\mathbb{E}M = I$. Inflation is excluded: a fixed factor $\tau \neq 1$ adds an $O(1)$ bias to $\mathbb{E}M$, which the expansion does not cover; we do not analyse factors that approach one as N grows. N_0 is not explicit. Uniformity is essential, and it holds because condition (iii) of (6.2) bounds every admissible E by the same $\|S - I\|$.

8 Prior-normalised loss at finite ensemble size

Return to a scalar observation in dimension $m \geq 2$ and take $W = I_m$ in canonical coordinates, that is, Mahalanobis loss with weight P^{-1} in physical variables. Let $f(Q)$ be the risk of complete localisation in the orthonormal basis Q, and let $\mathrm{Stab} = \{Q \in O(m) : Qe_1 = e_1\}$. Every $Q \in \mathrm{Stab}$ is a canonical basis that differs only in how the unobserved block is completed.

Proposition 8.1 (Second variation under $W = I$)**.** *Let $\zeta_{\nu,a,\beta} = 2\beta\big[(a\beta+1)\omega_{\nu,\beta} - 2\kappa_{\nu,a,\beta}\big]$, with κ, ω as in Proposition 4.1.*

1. *$f(VQ) = f(Q)$ for all $V \in \mathrm{Stab}$ and $Q \in O(m)$.*
2. *For $\Omega = \sum_{l\geq 2} \xi_l (e_l e_1^\mathsf{T} - e_1 e_l^\mathsf{T}) + \Omega_u$, with Ω_u skew and supported on the unobserved block, $\frac{d^2}{dt^2} f(e^{t\Omega})|_{t=0} = \zeta_{\nu,a,\beta}\,|\xi|^2$.*
3. *If $\zeta_{\nu,a,\beta} > 0$, equivalently $2\kappa_{\nu,a,\beta} < (a\beta+1)\omega_{\nu,\beta}$, there is a neighbourhood $\mathcal{N}$ of I in $O(m)$ with $f(Q) \geq f(I)$ on $\mathcal{N}$ and equality exactly on $\mathcal{N} \cap \mathrm{Stab}$: canonical complete localisation is a strict local minimiser up to the irrelevant completion. If $\zeta_{\nu,a,\beta} < 0$, it is not a local minimiser. If $\zeta_{\nu,a,\beta} = 0$, the proposition decides nothing.*

Proof. (1) For $V \in \mathrm{Stab}$ and any M, $k(VMV^\mathsf{T}) = Vk(M)$ because $V^\mathsf{T} e_1 = e_1$. With $W = I$, $r(k) = am - 2ahe_1^\mathsf{T} k + (a\beta+1)|k|^2$ is invariant under $k \mapsto Vk$. Complete localisation in VQ gives $V\,[Q\,\mathrm{diag}(Q^\mathsf{T} S' Q) Q^\mathsf{T}]\,V^\mathsf{T}$ with $S' = V^\mathsf{T} S V$, which has the law of S; the sample mean is unaffected. Hence $f(VQ) = f(Q)$.

(2) By the coupling, $f(e^{t\Omega}) = \mathbb{E}\, r(k(M_t))$ with $M_t = e^{t\Omega}\Lambda e^{-t\Omega} = \Lambda + tX + \frac{t^2}{2}[\Omega, X] + O(t^3)$ and $X = [\Omega, \Lambda]$. Direct computation gives $X_{l1} = \xi_l(D_1 - D_l)$ for $l \geq 2$ and $X_{11} = 0$, hence

$$(M_t)_{11} = D_1 - t^2 \sum_{l\geq 2} \xi_l^2 (D_1 - D_l) + O(t^3), \qquad (M_t)_{l1} = t\, \xi_l (D_1 - D_l) + O(t^2).$$

Ω_u enters only the $O(t^2)$ part of $(M_t)_{l1}$. Now $r = a(m-1) + \psi((M_t)_{11}) + \beta(a\beta+1)\sum_{l\geq 2}(M_t)_{l1}^2(1+\beta(M_t)_{11})^{-2}$ with $\psi(u) = (a+\beta u^2)(1+\beta u)^{-2}$, and the $O(t^2)$ part of $(M_t)_{l1}$ contributes only at order t^3. Using $\psi'(u) = 2\beta(u-a)(1+\beta u)^{-3}$, $\mathbb{E}[D_1 - D_l \mid D_1] = D_1 - 1$ and $\mathbb{E}[(D_1 - D_l)^2 \mid D_1] = (D_1 - 1)^2 + 2/\nu$, the second derivative is $-4\beta\kappa|\xi|^2 + 2\beta(a\beta+1)\omega|\xi|^2$. Polynomial domination justifies differentiating under $\mathbb{E}$.

(3) The map $(V, \xi) \mapsto V \exp(\sum_l \xi_l (e_l e_1^{\mathsf{T}} - e_1 e_l^{\mathsf{T}}))$ from Stab $\times\, \mathbb{R}^{m-1}$ to $O(m)$ has bijective differential at $(I, 0)$, since the Lie algebra of $O(m)$ is the direct sum of the unobserved block and the $(1, l)$ generators; it is a local diffeomorphism. By (1), $f(Ve^{\Omega_\xi}) = \phi(\xi) := f(e^{\Omega_\xi})$. By Theorem 3.3 with $W_{1l} = 0$, $\nabla\phi(0) = 0$; by (2) with $\Omega_u = 0$ and polarisation, $\nabla^2\phi(0) = \zeta I_{m-1}$; and f is C^3 by polynomial domination. If $\zeta > 0$, $\phi(\xi) > \phi(0)$ for small $\xi \neq 0$, and $Ve^{\Omega_\xi} e_1 = e_1$ only if $\xi = 0$. If $\zeta < 0$, ϕ decreases along every direction. □

Open problem 8.2 (The inequality $\zeta_{\nu,a,\beta} > 0$). Prove that $2\kappa_{\nu,a,\beta} < (a\beta+1)\omega_{\nu,\beta}$ for all $\nu \geq 1$, $\beta > 0$ and $a \in \{1, 1 + 1/N\}$.

Deterministic quadrature over fourteen ensemble sizes between 2 and 1000, 31 log-spaced $\beta \in [10^{-3}, 10^3]$ and both mean conventions gave the ratio $(a\beta+1)\omega/(2\kappa) > 1$ everywhere. Its minimum was 1.0020, at $N = 1000$, $\beta = 10^{-3}$. The ratio tends to 1^+ as $\beta \to 0$ and to $(1+\beta)^2$ as $N \to \infty$, so any proof must be tight in the weak-observation limit. Stein's identity for the gamma law, $\mathbb{E}[(D-1)g(D)] = (2/\nu)\mathbb{E}[Dg'(D)]$, removes the factors $D-1$ but leaves an integrand that is not pointwise positive. Numerically, then, for a scalar observation, canonical complete localisation is a strict local minimiser modulo completion under prior-normalised loss at every tested N; this is the finite-N counterpart of the leading-order result. Global finite-N optimality under $W = I$, and the several-observation analogue, remain open.

9 A constructive two-dimensional rotation and its size

In dimension two put $B = \mathrm{diag}(\mu_1, \mu_2)$ and let A have entries A_{11}, A_{12}, A_{22}. Expanding F for the rotation by θ gives

$$F(\theta) = \bar{F} + f_c \cos(4\theta) + f_s \sin(4\theta),$$

where

$$f_c = \frac{(\mu_1 - \mu_2)(A_{11} - A_{22})}{4}, \qquad f_s = \frac{(\mu_1 - \mu_2)A_{12}}{2}, \qquad \bar{F} = \frac{\mathrm{tr}\, A\ \mathrm{tr}\, B}{2} + f_c.$$

Hence $\min_\theta F = \bar{F} - (f_c^2 + f_s^2)^{1/2}$, attained where $4\theta = \mathrm{atan2}(f_s, f_c) + \pi \pmod{2\pi}$; if $f_c = f_s = 0$, all angles tie at this order.

For the example (4.1), $C = \mathrm{diag}(1/2, 1)$, $B = \mathrm{diag}(1/2, 0)$ and $A = \left(\begin{smallmatrix} 1/2 & 1/2 \\ 1/2 & 1 \end{smallmatrix}\right)$, so

$$F(\theta) = \frac{5 - \cos(4\theta) + 2\sin(4\theta)}{16}, \qquad \theta_* = -\tfrac{1}{4}\arctan 2 \simeq -15.8587^\circ.$$

The minimum is $(5 - \sqrt{5})/16$ against $F(0) = 1/4$, a reduction of $(\sqrt{5} - 1)/4 = 30.90\%$ in the leading covariance-estimation term. The leading sample-mean term is $3/(2N)$, so the reduction in the leading *excess over the Bayes risk* is $(\sqrt{5} - 1)/16 = 7.73\%$. The Bayes risk is 2, so neither figure is a reduction of the whole risk.

For a reproducible finite-angle illustration we use the rational rotation

$$Q_{\mathrm{num}} = \frac{1}{65}\begin{pmatrix} 63 & 16 \\ -16 & 63 \end{pmatrix}, \qquad \theta_{\mathrm{num}} = -2\arctan(1/8) \simeq -14.2500^\circ, \tag{9.1}$$

chosen for its simple rational entries rather than its optimality; it differs from the tiny certificate rotation of (4.2). Table 3 reports deterministic numerical integration, including the ensemble mean, and the same values divided by the Kalman-filter (Bayes) risk, the normalisation of Snyder and Hakim's figures. These values are illustrations, not interval certificates. The numerically optimal angle at each N lies between -14.1° ($N = 2$) and -16.5° ($N = 5$) and approaches θ_* as N grows.

Table 3: Physical squared risk for (4.1) with the ensemble mean, canonical basis against the fixed rotation Q_{num}. The Bayes (Kalman-filter) risk is 2. "Excess" is risk minus 2.

	Risk		Risk / Kalman-filter risk		Reduction of	
N	Canonical	Rotated	Canonical	Rotated	total risk	excess
2	3.377282	3.250331	1.6886	1.6252	3.759%	9.217%
3	2.831656	2.746111	1.4158	1.3731	3.021%	10.286%
5	2.459409	2.413463	1.2297	1.2067	1.868%	10.001%
10	2.214934	2.195319	1.1075	1.0977	0.886%	9.126%
20	2.103745	2.094968	1.0519	1.0475	0.417%	8.460%
50	2.040600	2.037358	1.0203	1.0187	0.159%	7.985%
100	2.020150	2.018575	1.0101	1.0093	0.078%	7.814%
200	2.010037	2.009262	1.0050	1.0046	0.039%	7.727%

The total-risk gain shrinks like $1/N$ because every method approaches the Bayes risk, but the relative gain in excess risk stays near 8%. The next section shows that in a realistic configuration the excess-risk gain is far larger.

10 Snyder and Hakim's one-dimensional testbed, reconstructed

Snyder and Hakim's evidence for their conjecture comes from a one-dimensional example [27, Section 4, Figs 3 and 5]. We rebuild that setting from their text and evaluate it with the objective F of Theorem 6.1 and by Monte Carlo. This is a *reconstruction*, not a replication of their figures: their code is not available to us, the correlation scale is partly illegible in the published text, and several details (grid placement, observation positions, boundary treatment) had to be inferred. Table 4 lists the choices; Section 10.4 reports the sensitivity to them.

Table 4: Reconstructed setting. "SH" marks a value stated in [27, Section 4]; "ours" marks a choice we made where the text leaves it open.

Item	Value	Source
State	$N_x = 200$, $x_i = (i + \frac{1}{2})/200$ on $(0, 1)$, not periodic	SH; placement ours
Prior	$x \sim N(0, P)$, unit variance, Matérn $\nu = 5/2$	SH
Correlation	$(1 + s + s^2/3)e^{-s}$, $s = \sqrt{5}\, r/\rho$, $\rho = 0.03646$	SH form; ρ ours
	(correlation exactly $1/2$ at $r = 0.038$)	SH: $1/2$ at $r = 0.038$
Observations	$N_y = 50$ point values, every fourth grid point	SH; offset ours
Observation error	independent, unit variance ($R_{\mathrm{o}} = I$)	SH
Loss	physical squared error, divided by Kalman-filter MSE	SH Figs 3, 5
Localisation	complete (diagonal) in each whitening basis	SH Section 3.2

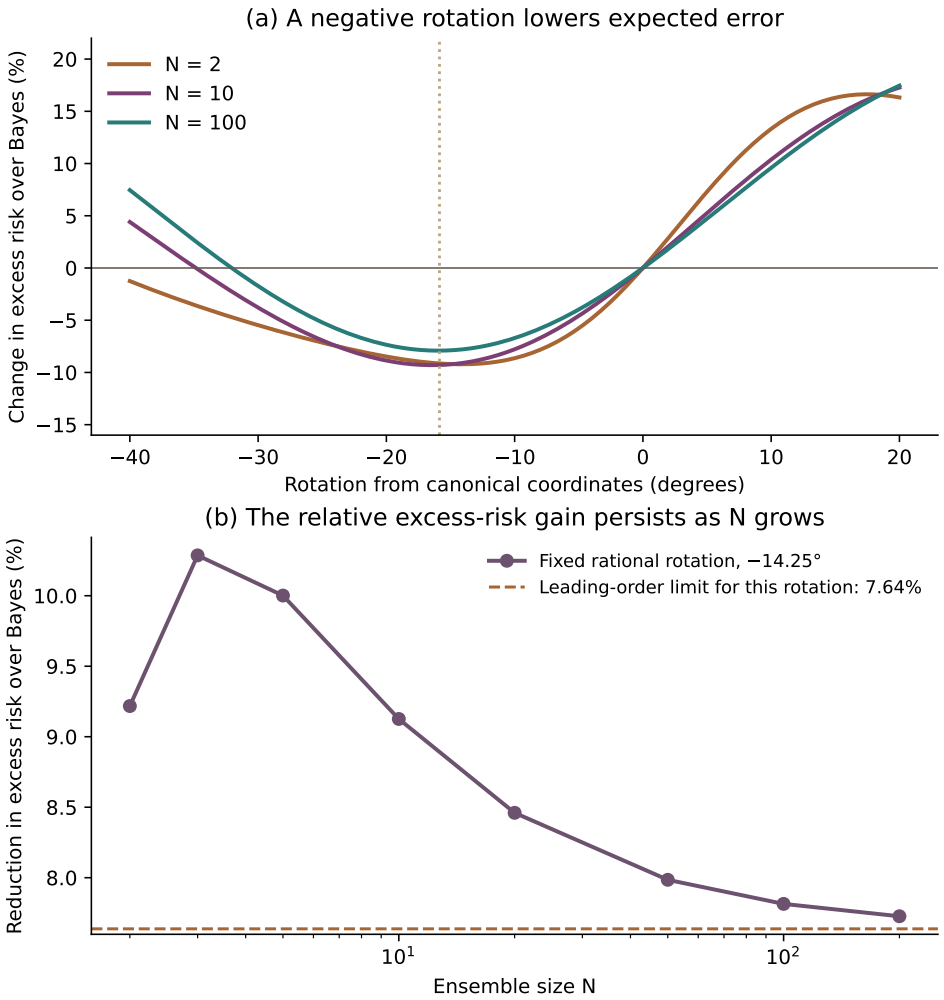


Figure 1: Deterministic integration for the example (4.1), ensemble-mean update. (a) Change in excess risk over the Bayes risk, relative to the canonical basis, as a function of the rotation angle, for $N = 2$, 10 and 100; negative values favour the rotation. The dotted line marks the leading-order optimum θ_*. (b) Reduction in excess risk for the fixed rotation Q_{num} against N; the dashed line is the leading-order limit for Q_{num}, 7.64% (the leading-order optimum θ_* would give 7.73%). The rotated column of Table 3 uses the same rotation.

In canonical coordinates $x = Lz$ with $L = P^{1/2}V$, where V holds the right singular vectors of $R_o^{-1/2}HP^{1/2}$; the loss weight is $W = L^\mathsf{T}L = V^\mathsf{T}PV$. With C, B and A as in Section 6, every whitening basis is an orthogonal matrix U acting on z. We compare four bases:

- *canonical*: $U = I$, Snyder and Hakim's Eq. 8;
- *P-eigen*: the eigenvectors of P, their Eq. 15, which ignore the observations;
- *ZCA*: symmetric whitening $P^{-1/2}x$ [14], which keeps each whitened variable attached to one grid point;
- *F-optimised*: the best basis found by Riemannian gradient descent on $O(200)$ with a Cayley retraction, started from both the canonical and the ZCA basis. It is not certified to be a global minimiser of F.

The two descents end within 6×10^{-5} of each other in F. After 4000 iterations neither met the strict gradient tolerance (final Riemannian gradient norm about 1.7×10^{-4}), so the reported F_{opt} is an upper bound on the minimum; its last 1000 iterations changed F by less than 0.02%.

10.1 The leading-order objective

The canonical basis fails the criterion of Theorem 7.1 on both counts. The commutator has $\|[W, C]\|_F / \|WC\|_F = 0.056$. And, even ignoring the commutator, the diagonal weights are ordered the wrong way: across canonical directions the loss weights $\lambda_i = u_i^\mathsf{T}Au_i$ and the information weights $\mu_i = u_i^\mathsf{T}Bu_i$ have correlation $+0.88$, whereas (7.1) asks for an opposite ordering. Table 5 gives F.

Table 5: Leading objective $F(U)$ in the reconstructed testbed. "Spread" is $(\sum_i \mu_i)^2 / \sum_i \mu_i^2$, the effective number of basis directions that carry observation information ($\sum_i \mu_i = \operatorname{tr} B = 15.92$ for every basis).

Basis	$F(U)$	Spread	$\max_i \mu_i$
Canonical (SH Eq. 8)	12.004	27.3	0.81
P-eigen (SH Eq. 15)	12.030	27.3	0.81
Haar-random, exact mean $(\operatorname{tr} A \operatorname{tr} B + 2\operatorname{tr}(AB))/(m+2)$	2.229		
sampled mean (40 draws)	2.229 ± 0.002		
$\operatorname{tr} A \operatorname{tr} B/m$	2.131		
ZCA (symmetric whitening)	2.134	198.4	0.13
F-optimised	1.963	185.6	0.12

At leading order the canonical basis is close to the worst choice. A Haar-random orthogonal basis has an F more than five times smaller, and the best value found is 84% below canonical (so the true minimum is at least that far below). The P-eigen basis is almost indistinguishable from canonical, in agreement with Snyder and Hakim's Fig. 5. The ZCA basis, which they did not test, is within 9% of the best value found.

10.2 Finite-ensemble risk

For a fixed orthonormal U, complete localisation uses only the sample variances of $w = U^\mathsf{T}z$. By rotation invariance (Section 2) these are independent χ^2_ν/ν variables, independent of the ensemble mean, so the analysis-error covariance given them is exactly $a(I - KG)(I - KG)^\mathsf{T} + KK^\mathsf{T}$ with $K = M_U G^\mathsf{T}(I + GM_U G^\mathsf{T})^{-1}$. We average this exact conditional risk over 4000 draws per N, using the same draws for every basis so that paired differences are precise. Table 6 and Figure 2 report MSE divided by the Kalman-filter MSE ($\operatorname{tr}(WC) = 63.88$), as in Snyder and Hakim's figures.

The F-optimised basis removes 54% of the canonical basis's excess error over the Kalman filter at $N = 8$, 47% at $N = 16$ and 42% at $N = 64$. In Snyder and Hakim's units that is 7.4% of

Table 6: MSE divided by Kalman-filter MSE, ensemble-mean update. Standard errors are at most 2 in the last digit for canonical and P-eigen at $N = 4$, smaller elsewhere. Brackets: the leading-order prediction $1 + [\mathrm{tr}(WC^2)/N + 2F/(N-1)]/\mathrm{tr}(WC)$. Last column: paired difference, F-optimised minus canonical, $\pm$ one standard error.

N	Canonical	P-eigen	ZCA	F-optimised	Difference
4	1.366 [1.230]	1.367	1.135	1.134 [1.125]	-0.2324 ± 0.0018
8	1.137 [1.106]	1.137	1.064	1.063 [1.061]	-0.0743 ± 0.0006
16	1.058 [1.051]	1.058	1.031	1.031 [1.030]	-0.0276 ± 0.0002
32	1.027 [1.025]	1.027	1.015	1.015 [1.015]	-0.0116 ± 0.0001
64	1.013 [1.013]	1.013	1.008	1.008 [1.008]	-0.0054 ± 0.0000

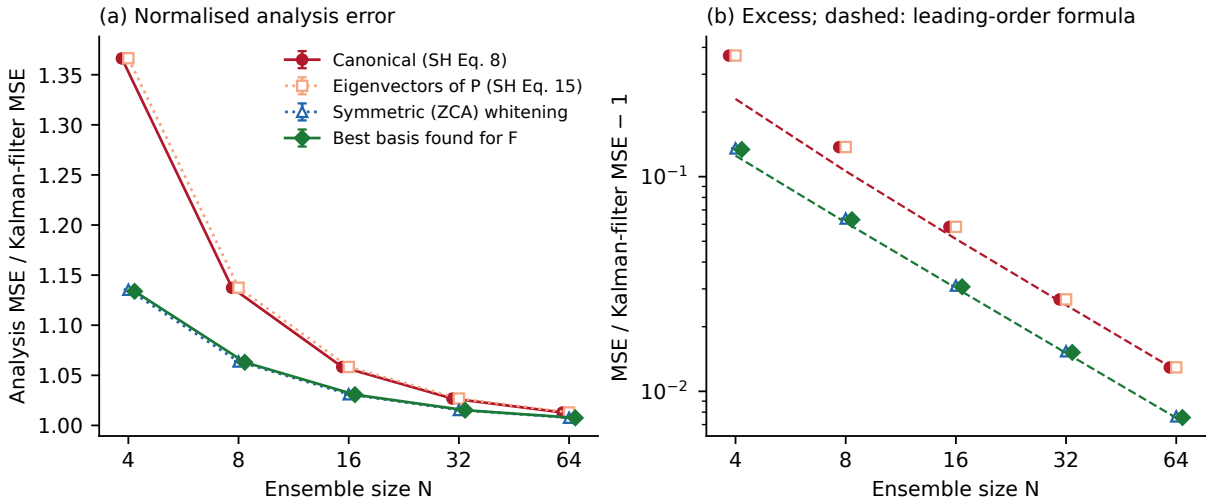


Figure 2: Reconstructed Snyder–Hakim testbed: analysis MSE divided by Kalman-filter MSE for complete localisation in four whitening bases; one analysis, oracle prior covariance. (a) Normalised MSE; error bars are ± 2 standard errors, mostly smaller than the markers. P-eigen overlies canonical, and ZCA overlies the F-optimised basis. (b) The same results as excess over the Kalman filter on log axes; dashed lines are the leading-order formula of Theorem 6.1.

Kalman-filter MSE at $N = 8$ and 2.8% at $N = 16$. The gap between canonical and P-eigen is $+0.01\%$ of Kalman-filter MSE, with a paired standard error of 2×10^{-6}. The known-mean update gives the same picture: 1.071 against 1.009 at $N = 8$. The leading-order formula is accurate for the spread-out bases from $N = 8$, and for canonical from $N = 32$. At small N it understates the canonical risk, so the leading order is conservative about the size of the failure.

Equivalent ensemble size. Interpolating the F-optimised risk curve, its risk at N_{eq} matches canonical at N for $(N, N_{\mathrm{eq}}) = (8, 3.9)$, $(16, 8.6)$, $(32, 18.3)$ and $(64, 37.5)$; the ZCA values are within 0.5 of these. In this setting canonical localisation needs roughly 1.7 to 2 times as many members as symmetric whitening to reach the same analysis error.

Mechanism. The canonical basis concentrates all observation information into the 50 observed directions: its μ_i range up to 0.81, with an effective spread of 27 directions. The same directions carry most of the loss weight λ_i. Each localised variance has relative sampling error $\sqrt{2/\nu}$, and Theorem 6.1 prices that error at $\lambda_i \mu_i$, so putting large λ and large μ in the same directions maximises the cost. The F-optimised basis spreads the same total information $\mathrm{tr}\, B$ over about 186 directions, none with $\mu_i > 0.12$, and reverses the ordering: the correlation of λ_i and μ_i becomes -0.96. Each optimised vector keeps between 12% and 61% of its weight in the observed canonical block (median 19%), so it mixes observed with unobserved canonical directions, as Theorem 7.1 and Corollary 5.2 suggest. Symmetric whitening achieves almost the same spreading without any optimisation, because each of its variables is tied to one grid point. Any basis not

aligned with the canonical directions does nearly as well: a single Haar-random basis gives 1.064 at $N = 8$.

Prior-normalised loss. With $W = I$ (Mahalanobis loss in x) the same testbed behaves as Theorem 7.1 predicts. The canonical basis is stationary for F, with a gradient that is exactly zero. Descents started from ZCA, P-eigen and two Haar-random bases all end at or above canonical: the best reaches 3.60869, against $F_{\text{can}} = 3.60866$. The Monte Carlo agrees. ZCA is now worse than canonical by 0.0159 ($N = 8$) and 0.0037 ($N = 32$) of the Kalman-filter risk, and P-eigen is worse by 5×10^{-5}. The choice of loss, not the testbed, decides which basis is better.

Estimated bases. The bases above use the true P. To see how much of the gain survives estimation, we built each basis from an independent pilot sample of size N_p, with the pilot covariance Gaspari–Cohn localised and a ridge δI added. The analysis ensembles were then simulated in physical space, because estimated whitening leaves the transformed variables correlated and the rotation-invariance shortcut no longer applies. At $N = 8$ the oracle values in this physical-space simulation (800 replicates; compare Table 6) are 1.136 (canonical), 1.064 (ZCA) and 1.063 (F-optimised). One pilot draw per configuration gives:

- $N_p = 200$, GC half-width 0.2, $\delta = 10^{-6}$: estimated ZCA 1.068 and estimated F-optimised 1.067. These keep 93% and 95% of the oracle gain at $N = 8$, and 70% and 72% at $N = 16$.
- $N_p = 200$, GC half-width 0.2, $\delta = 10^{-4}$: estimated ZCA and F-optimised keep 53% and 37% of the oracle gain at $N = 8$; at $N = 16$ ZCA keeps 8% and estimated F-optimised is *worse* than oracle canonical (−28% of the gain). So $\delta = 10^{-6}$ is the best of the $N_p = 200$ settings tried, not the only one.
- $N_p = 50$: the best regularisation tried ($\delta = 10^{-4}$) keeps about 60% at $N = 8$ but none at $N = 16$. With $\delta = 10^{-6}$ both spread-out bases are worse than oracle canonical.
- Heavier regularisation (half-width 0.1, $\delta = 10^{-3}$): estimated ZCA and F-optimised are 0.12 to 0.26 of Kalman-filter MSE *worse* than canonical at both sample sizes.
- An unregularised pilot with $N_p = 2000$ recovers essentially all of the gain (estimated ZCA 1.064, F-optimised 1.063 at $N = 8$).
- Estimated *canonical* is robust: within 0.008 ($N_p = 200$) and 0.031 ($N_p = 50$) of oracle canonical at $N = 8$, for every regularisation tried.

Random estimation error in the whitening is harmless, since nearly any basis not aligned with canonical does well. Systematic distortion is not. About 100 eigenvalues of P lie below 10^{-3} (the smallest is 2.9×10^{-5}), and a ridge or taper that inflates them leaves the spread-out coordinates strongly correlated, so complete localisation discards real covariance. Canonical localisation depends only on the 50 observed, large-scale directions, which are well estimated. In this testbed the oracle gain is real but fragile in practice. Turning it into a scheme needs a whitening that is accurate in the small-variance directions, or a basis chosen to be robust to their misestimation; this matches Snyder and Hakim's remark that the optimal transformation does not lead directly to practical schemes.

10.3 The homogeneous (Fourier) case

Snyder and Hakim single out the homogeneous case, with uniform observations and homogeneous errors, as the one where the optimal transformation is known analytically: the discrete Fourier transform [27, Section 7]. Under physical loss Theorem 7.1 shows that it is generally not leading-order optimal.

Corollary 10.1 (Fourier bases under physical loss)**.** *Let P be circulant with mode variances $p_i > 0$, $H = I$ and $R_{\text{o}} = \sigma^2 I$, $\sigma > 0$. The Fourier basis is canonical, with $W = \operatorname{diag}(p_i)$, $c_i = \sigma^2/(\sigma^2 + p_i)$, $\lambda_i = c_i^2 p_i = \sigma^4 p_i/(\sigma^2 + p_i)^2$ and $\mu_i = p_i/(\sigma^2 + p_i)$. If two modes satisfy $p_j < p_i \le \sigma^2$, the Fourier basis does not minimise F: a 45° mixing of those two modes lowers F by $(\lambda_i - \lambda_j)(\mu_i - \mu_j)/2 > 0$.*

Proof. Here W and C are both diagonal in the Fourier basis, so they commute. The maps $p \mapsto \mu$ and $p \mapsto \lambda$ are both strictly increasing on $(0, \sigma^2]$, since $\frac{d}{dp}\, p/(\sigma^2+p)^2 = (\sigma^2-p)/(\sigma^2+p)^3$. Hence $(\lambda_i - \lambda_j)(\mu_i - \mu_j) > 0$, violating (7.1), and the 45° reduction is the one computed in the proof of Theorem 7.1. □

The commutation condition holds here, and only the ordering fails, so the two conditions of Theorem 7.1 are independent. Numerically we used a periodic 64-point grid with the same Matérn correlation and $\sigma^2 = 1$. The real Fourier basis diagonalises P to 10^{-14}, and 47 of the 64 modes have $p_i < \sigma^2$. Every one of the 1058 distinct pairs below σ^2 violates the ordering, and so do 671 further pairs. The predicted and computed 45° reductions agree to 10^{-15}, and descent from a perturbed Fourier basis lowers F from 3.08 to 1.67. As a control we set σ^2 below every mode variance; then no pair violates the ordering and the Fourier basis is leading-order optimal, as the corollary requires. The same test applies to any spectral or wavelet-diagonal localisation: under physical loss it is leading-order optimal only if loss weight and information are oppositely ordered across its modes.

10.4 Limits of the reconstruction

- *Oracle basis.* The F-optimised and ZCA bases use the true P. The estimated-basis test shows that the gain survives only with careful, regularisation-dependent estimation, and it used one pilot draw per configuration.
- *One analysis, Gaussian ensemble.* There is no cycling, no model error and no inflation. Complete localisation is compared with complete localisation; tuned Gaspari–Cohn localisation in physical space is not part of this comparison.
- *Reconstruction uncertainty.* Our calibrated $\rho = 0.0365$ gives correlation $1/2$ at exactly $r = 0.038$; the two readings $\sqrt{2}/40$ and $1.5/40$ of the partly illegible scale give $r = 0.037$ and 0.039, inside our sensitivity range. Changing ρ by factors 0.5 and 1.5, or N_y to 25 or 100, leaves the conclusion unchanged: canonical F stays 4 to 8 times the best value found, and at $N = 8$ the F-optimised basis improves MSE/KF-MSE by 0.059 to 0.077 (all paired $z > 60$).
- *Comparison with Snyder and Hakim's Fig. 5.* They report canonical better than P-eigen by more than 0.5% of Kalman-filter MSE for $N \leq 16$; we find 0.01%. For one Gaussian realisation the relative standard deviation of the squared error at Kalman-filter accuracy is 0.27, so an average over 800 unpaired realisations has a standard error near 1% of Kalman-filter MSE. Differences of their size are within that sampling noise, but we cannot tell from the text whether their comparison was paired.
- *Optimiser.* F_{opt} is an upper bound on the minimum over $O(200)$; a better optimiser could only strengthen the contrast.

11 Discussion

11.1 What the results mean in practice

Population diagonalisation is not the same as estimation-risk minimisation. Canonical coordinates remain the right way to *describe* the Gaussian update: they decouple it exactly. Choosing a basis for *localisation* is a different problem, because it trades sampling error against the loss used to judge the analysis. In the reconstructed testbed, under physical squared error, the canonical basis concentrates observation information in the directions that also carry the most loss weight, which is exactly where sampling error in the variances does most damage. Physical loss alone does not imply this: Theorem 7.1 decides it, and Remark 7.2 gives a case where canonical coordinates are leading-order optimal under physical loss.

A loss-aware basis. The leading objective $F(U) = \sum_i (u_i^\mathsf{T} B u_i)(u_i^\mathsf{T} A u_i)$ uses the same ingredients as the canonical transformation (P, H, R) plus the loss. It gives a computable rule for choosing a complete-localisation basis, and Theorem 7.1 says when canonical coordinates are already optimal. A practical reading, when the prior covariance is accurately known, is: choose the whitening basis with the loss in mind, and check the criterion before relying on canonical coordinates.

Simple bases do well. In the reconstructed testbed, plain symmetric (ZCA) whitening, which needs no singular value decomposition and is cheaper than the canonical transformation, comes within 9% of the best F found and within 0.002 of the finite-N risk of the best basis found at every N tested. Its advantage over canonical localisation is equivalent to roughly doubling the ensemble. With the prior covariance known, the message for this testbed is that canonical coordinates are a poor choice before complete localisation and that no fine-tuned rotation is needed to do much better; with an estimated covariance the advantage is fragile (below).

When canonical coordinates are right. Under prior-normalised loss the canonical basis is leading-order optimal for every observation network, within the class stated after Theorem 7.1, and, for a scalar observation, a strict local minimum at finite N in every case we computed. If a system is judged by a Mahalanobis-type metric, Snyder and Hakim's choice is the right target.

Estimation is the main barrier. With an estimated prior covariance, the advantage of the spread-out bases survived only with accurate estimation of the small-variance directions; heavy regularisation reversed it. Canonical localisation was robust to estimation.

11.2 Limits

The theory concerns one analysis step with an independent Gaussian forecast ensemble and an oracle prior covariance used to build every basis. The comparison classes are deterministic tapers; sample-adaptive corrections [3], shrinkage, hybrid covariances and tuned distance-based localisation are outside them. The exact all-N taper comparison is for a scalar observation; with several observations the finite-N obstruction is against rotated complete localisations, and the comparison with all canonical tapers is asymptotic (Corollary 7.6). The expansion holds for fixed dimensions, whereas operational systems have $m \gg N$. No real observations, forecast model or cycling were tested, and the testbed is a reconstruction with the uncertainties listed in Section 10.4.

11.3 Open problems

1. *Prior-normalised loss at finite N.* Prove $\zeta_{\nu,a,\beta} > 0$ (Open problem 8.2), and decide whether canonical complete localisation is globally optimal under $W = I$ at finite N, for scalar and for vector observations.
2. *Two observed directions.* Prove Conjecture 5.4, which would make $[W, C] = 0$ necessary and sufficient for finite-N stationarity.
3. *Estimated bases.* Find a data-driven estimator of a loss-aware basis that is robust to misestimation of small prior variances, and quantify the loss relative to the oracle.
4. *Cycling.* Determine whether the one-step advantage persists, grows or disappears when the forecast covariance is itself propagated from an ensemble.
5. *Joint taper and rotation.* Optimise the basis and a partial (not complete) taper jointly, including inflation, rather than comparing complete localisations.
6. *Ties and higher order.* Characterise the N^{-2} term that decides the comparison when the leading objective ties (Remark 4.2), and prove the $O(\nu^{-2})$ remainder sketched in Remark 6.4.
7. *High dimension.* Develop the analogue of F in a proportional regime $m \sim N$.

Reproducibility and assurance

The proofs in this paper are written out in full but have not been machine-verified. Two kinds of computational evidence are kept separate. *Exact arithmetic* (`exact_certificates.py`, Python fractions) checks the finite constants used in the proofs: the polynomial certificate $460/773$, the curvature bounds $-56/773$ and $-508/773$, the second-derivative bound $H_0 = 23148$ and the rational rotation certificates. *Floating-point computations* (`numerical_checks.py`, `theory_checks.py`, `sh_testbed.py`) evaluate expectations by deterministic quadrature and by seeded Monte Carlo, including a Monte Carlo of the actual ensemble procedure with 4×10^6 ensembles that checks Lemma 3.1, Theorem 3.3, the inflation-robust derivative, Proposition 4.1 and two Table 3 entries within stated standard errors, together with negative controls that must fail. Floating-point results are illustrations and consistency checks, not interval certificates. Every script records its own SHA-256 hash, writes to a new output directory and exits with an error if any check fails. The paper received internal, AI-assisted review only; it has not been externally peer reviewed. The literature search behind the novelty statement was bounded and is dated 3 November 2026; it is not a certificate of priority.

This is version 0.1.0-candidate, an unrefereed research candidate. The paper, its LaTeX source, the scripts and their recorded outputs are archived at doi:10.5281/zenodo.23926884 (all versions: doi:10.5281/zenodo.23926883) and in the repository `https://github.com/ipitchford/canonical-covariance-localisation-failure`.

A An explicit rational finite-angle certificate

Work with (4.1), with either $a = 1$ or $a = 1 + 1/N$. The moments of $D \sim \chi^2_\nu/\nu$ give

$$\mathbb{E}(D-1)^2 = \frac{2}{\nu}, \quad \mathbb{E}[D(D-1)^2] = \frac{2}{\nu}\Big(1+\frac{4}{\nu}\Big), \quad \mathbb{E}[D^2(D-1)^2] = \frac{2}{\nu}\Big(1+\frac{14}{\nu}+\frac{24}{\nu^2}\Big).$$

Cauchy–Schwarz therefore yields

$$\mathbb{E}\frac{(D-1)^2}{(1+D)^2} \geq \frac{(\mathbb{E}(D-1)^2)^2}{\mathbb{E}[(D-1)^2(1+D)^2]} = \frac{\nu}{2\nu^2+11\nu+12},$$

and since $\gamma_{\nu,a,1} \geq \mathbb{E}[(D-1)^2(1+D)^{-2}]$ (proof of Theorem 3.3),

$$\mathcal{R}'_N(0) \geq g_\nu := \frac{2\nu}{2\nu^2+11\nu+12}.$$

For $\nu = 2$ and $\nu = 3$ this gives the same value $2/21$.

Next, a second-derivative bound valid for every angle. Put $\Sigma = D_1 + D_2$. For the rotated covariance and gain, $\|M_\theta\| \leq \Sigma$, $\|M'_\theta\| \leq \Sigma$, $\|M''_\theta\| \leq 2\Sigma$, and differentiating $k = M_\theta e_1/(1 + (M_\theta)_{11})$ gives $\|k\| \leq \Sigma$, $\|k'\| \leq \Sigma(1+\Sigma)$, $\|k''\| \leq 2\Sigma(1+\Sigma)^2$. Since $\|W\| = (3+\sqrt{5})/2 < 3$ and $a \leq 3/2$,

$$|\mathcal{R}''_N(\theta)| \leq 18\,\mathbb{E}[\Sigma(1+\Sigma)^2] + 45\,\mathbb{E}[\Sigma^2(1+\Sigma)^2].$$

Σ is gamma distributed with shape ν and scale $2/\nu$; for $\nu \geq 1$ its first four moments are at most $2, 8, 48, 384$. Thus $|\mathcal{R}''_N(\theta)| \leq 18 \cdot 66 + 45 \cdot 488 = 23148 =: H_0$ for all θ. Set

$$t_\nu = \frac{g_\nu}{4H_0}, \qquad Q_\nu = \frac{1}{1+t_\nu^2}\begin{pmatrix} 1-t_\nu^2 & 2t_\nu \\ -2t_\nu & 1-t_\nu^2 \end{pmatrix}.$$

Its entries are rational, and it rotates through $-\delta_\nu = -2\arctan t_\nu$ with $t_\nu \leq \delta_\nu \leq 2t_\nu$. Taylor's theorem gives

$$\mathcal{R}_N(Q_\nu) - \mathcal{R}_N(I) \leq -g_\nu\delta_\nu + \tfrac{1}{2}H_0\delta_\nu^2 \leq -\frac{g_\nu^2}{8H_0} < 0.$$

For $\nu = 2$ this is (4.2).

B Evaluating the expectations numerically

For (4.1), write $u = \cos^2\theta\, D_1 + \sin^2\theta\, D_2$, $v = \cos\theta \sin\theta\, (D_1 - D_2)$ and $k = (u, v)^{\mathsf{T}}/(1 + u)$. The direct calculation integrates (2.3) over independent gamma variables D_1, D_2 (shape $\nu/2$, scale $2/\nu$) with a tensor generalised Gauss–Laguerre rule. The separate Laplace calculation uses $(1+u)^{-1} = \int_0^\infty e^{-t} e^{-tu}\, dt$ and $(1+u)^{-2} = \int_0^\infty t e^{-t} e^{-tu}\, dt$. With $k_\nu = \nu/2$, $r_1 = (1+t\cos^2\theta/k_\nu)^{-1}$, $r_2 = (1 + t\sin^2\theta/k_\nu)^{-1}$ and $\phi(t) = (r_1 r_2)^{k_\nu}$,

$$\mathbb{E}[e^{-tu}] = \phi(t), \qquad \mathbb{E}[(D_1 - D_2)e^{-tu}] = \phi(t)(r_1 - r_2),$$
$$\mathbb{E}[(D_1 - D_2)^2 e^{-tu}] = \phi(t)\Big[(r_1 - r_2)^2 + \frac{r_1^2 + r_2^2}{k_\nu}\Big],$$

so every first and second moment of the gain reduces to one-dimensional integrals. The two methods agree to within 5×10^{-14} over 32 comparisons. Both rest on (2.3) and (3.2), so their agreement checks the quadrature and its implementation; the model itself is checked by the Monte Carlo of the real ensemble procedure described in the reproducibility statement. The analytic theorems do not depend on any quadrature tolerance.

References

[1] Omar Al-Ghattas and Daniel Sanz-Alonso. Non-asymptotic analysis of ensemble Kalman updates: effective dimension and localization. *Information and Inference: A Journal of the IMA*, 13(1):iaad043, 2024. doi: 10.1093/imaiai/iaad043.

[2] Jeffrey L. Anderson. Exploring the need for localization in ensemble data assimilation using a hierarchical ensemble filter. *Physica D: Nonlinear Phenomena*, 230(1-2):99–111, 2007. doi: 10.1016/j.physd.2006.02.011.

[3] Jeffrey L. Anderson. Localization and Sampling Error Correction in Ensemble Kalman Filter Data Assimilation. *Monthly Weather Review*, 140(7):2359–2371, 2012. doi: 10.1175/MWR-D-11-00013.1.

[4] Rebecca Susanne Atkinson, Jonathan Flowerdew, Sue Hughes, and Ian Roulstone. Comparison of Traditional and Hybrid Forms of Optimal Localisation for Mitigation of Sampling Error in Ensemble Kalman Filters. *Tellus A: Dynamic Meteorology and Oceanography*, 76 (1):57–73, 2024. doi: 10.16993/tellusa.35.

[5] Jean-Christophe Bourin. Some inequalities for norms on matrices and operators. *Linear Algebra and its Applications*, 292(1-3):139–154, 1999. doi: 10.1016/S0024-3795(99)00024-5.

[6] Mark Buehner and Martin Charron. Spectral and spatial localization of background-error correlations for data assimilation. *Quarterly Journal of the Royal Meteorological Society*, 133(624):615–630, 2007. doi: 10.1002/qj.50.

[7] Jonathan Flowerdew. Towards a theory of optimal localisation. *Tellus A: Dynamic Meteorology and Oceanography*, 67(1):25257, 2015. doi: 10.3402/tellusa.v67.25257.

[8] Jun Ichi Fujii. A trace inequality arising from quantum information theory. *Linear Algebra and its Applications*, 400:141–146, 2005. doi: 10.1016/j.laa.2004.11.009.

[9] Reinhard Furrer and Thomas Bengtsson. Estimation of high-dimensional prior and posterior covariance matrices in Kalman filter variants. *Journal of Multivariate Analysis*, 98(2): 227–255, 2007. doi: 10.1016/j.jmva.2006.08.003.

[10] Daniel Hodyss and Matthias Morzfeld. How Sampling Errors in Covariance Estimates Cause Bias in the Kalman Gain and Impact Ensemble Data Assimilation. *Monthly Weather Review*, 151(9):2413–2426, 2023. doi: 10.1175/MWR-D-23-0029.1.

[11] Roger A. Horn and Charles R. Johnson. *Topics in Matrix Analysis*. Cambridge University Press, 1991. doi: 10.1017/CBO9780511840371.

[12] P. L. Houtekamer and Fuqing Zhang. Review of the Ensemble Kalman Filter for Atmospheric Data Assimilation. *Monthly Weather Review*, 144(12):4489–4532, 2016. doi: 10.1175/MWR-D-15-0440.1.

[13] Jeffrey D. Kepert. Covariance localisation and balance in an Ensemble Kalman Filter. *Quarterly Journal of the Royal Meteorological Society*, 135(642):1157–1176, 2009. doi: 10.1002/qj.443.

[14] Agnan Kessy, Alex Lewin, and Korbinian Strimmer. Optimal Whitening and Decorrelation. *The American Statistician*, 72(4):309–314, 2018. doi: 10.1080/00031305.2016.1277159.

[15] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004. doi: 10.1016/S0047-259X(03)00096-4.

[16] Andrew C. Lorenc. Improving ensemble covariances in hybrid variational data assimilation without increasing ensemble size. *Quarterly Journal of the Royal Meteorological Society*, 143 (703):1062–1072, 2017. doi: 10.1002/qj.2990.

[17] Benjamin Ménétrier and Thomas Auligné. Optimized Localization and Hybridization to Filter Ensemble-Based Covariances. *Monthly Weather Review*, 143(10):3931–3947, 2015. doi: 10.1175/MWR-D-15-0057.1.

[18] Benjamin Ménétrier, Thibaut Montmerle, Yann Michel, and Loïk Berre. Linear Filtering of Sample Covariances for Ensemble-Based Data Assimilation. Part I: Optimality Criteria and Application to Variance Filtering and Covariance Localization. *Monthly Weather Review*, 143(5):1622–1643, 2015. doi: 10.1175/MWR-D-14-00157.1.

[19] Benjamin Ménétrier, Thibaut Montmerle, Yann Michel, and Loïk Berre. Linear Filtering of Sample Covariances for Ensemble-Based Data Assimilation. Part II: Application to a Convective-Scale NWP Model. *Monthly Weather Review*, 143(5):1644–1664, 2015. doi: 10.1175/MWR-D-14-00156.1.

[20] Matthias Morzfeld and Daniel Hodyss. A Theory for Why Even Simple Covariance Localization Is So Useful in Ensemble Data Assimilation. *Monthly Weather Review*, 151(3):717–736, 2023. doi: 10.1175/MWR-D-22-0255.1.

[21] Cyrus Mostajeran and Rodolphe Sepulchre. Ordering positive definite matrices. *Information Geometry*, 1(2):287–313, 2018. doi: 10.1007/s41884-018-0003-7.

[22] Aamir Nadeem and Roland Potthast. Transformed and generalized localization for ensemble methods in data assimilation. *Mathematical Methods in the Applied Sciences*, 39(4):619–634, 2016. doi: 10.1002/mma.3496.

[23] África Periáñez, Hendrik Reich, and Roland Potthast. Optimal Localization for Ensemble Kalman Filter Systems. *Journal of the Meteorological Society of Japan. Ser. II*, 92(6): 585–597, 2014. doi: 10.2151/jmsj.2014-605.

[24] William Sacher and Peter Bartello. Sampling Errors in Ensemble Kalman Filtering. Part I: Theory. *Monthly Weather Review*, 136(8):3035–3049, 2008. doi: 10.1175/2007MWR2323.1.

[25] Anna Shlyaeva, Jeffrey S. Whitaker, and Chris Snyder. Model-Space Localization in Serial Ensemble Filters. *Journal of Advances in Modeling Earth Systems*, 11(6):1627–1636, 2019. doi: 10.1029/2018MS001514.

[26] Chris Snyder. Sampling error in the ensemble Kalman filter for small ensembles and high-dimensional states. Conference abstract EGU25-14641, Copernicus GmbH, 2025. doi: 10.5194/egusphere-egu25-14641.

[27] Chris Snyder and Gregory J. Hakim. An Optimal Linear Transformation for Data Assimilation. *Journal of Advances in Modeling Earth Systems*, 14(6):e2021MS002937, 2022. doi: 10.1029/2021MS002937.

[28] Peter Jan van Leeuwen. Comment on "Data Assimilation Using an Ensemble Kalman Filter Technique". *Monthly Weather Review*, 127(6):1374–1377, 1999. doi: 10.1175/1520-0493(1999)127<1374:CODAUA>2.0.CO;2.